



FAPS

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Production of Power Electronics for Controlling and Switching

Vorlesung Produktion elektrischer Motoren und Maschinen
Lecture Production of electric motors and machines

Power electronics must be adapted to the electric drives just like the control and cooling systems.

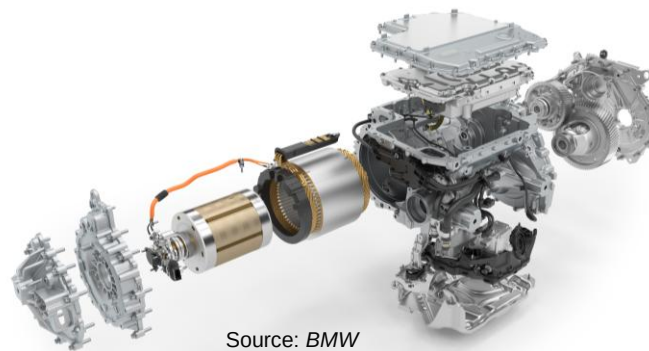
The efficiency of an electric drive system depends not only on the electric machine but also on the employed power electronics and their control. Therefore, it is essential to have an overview of the characteristics and the manufacturing of power electronic modules for electric machines.

After visiting the lecture unit, "Product design of power electronics for controlling and regulating", you should:

- Know the most important assembly and interconnection technologies for power electronics,
- Be able to differentiate between various package and module variants,
- Know the tasks of assembly and interconnection technology and power electronics in a power electronic system.



Source: Danfoss



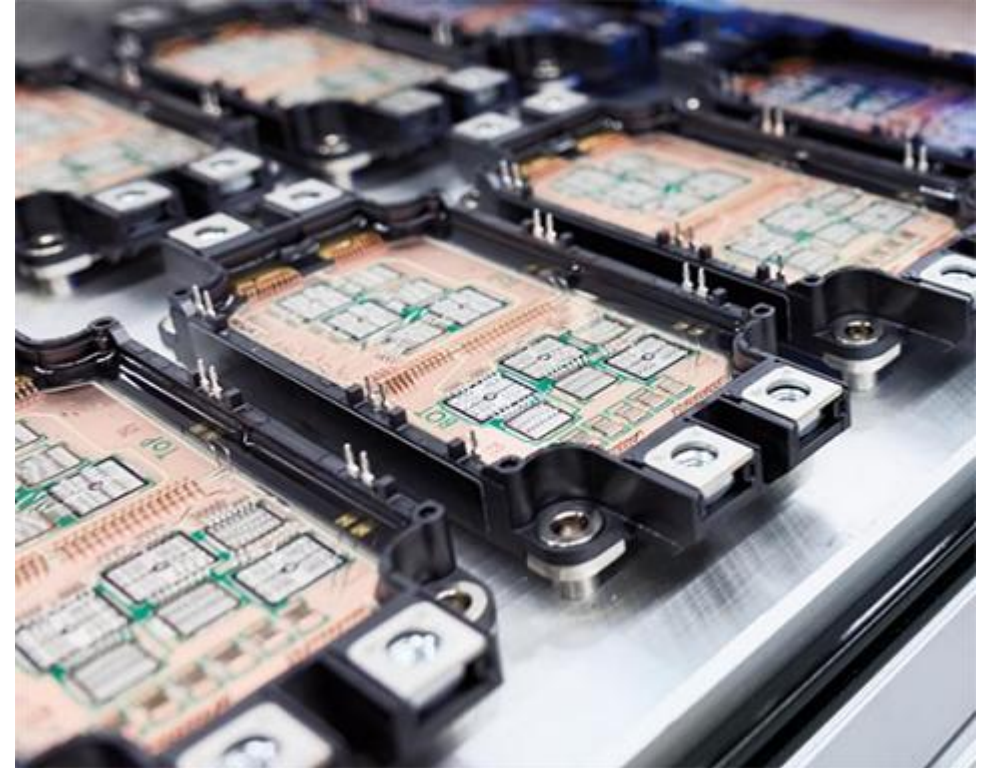
Source: BMW



Source: Marelli

The advanced unit 3 includes an overview of power electronics production for controlling and switching in electric drives.

- **Introduction to power electronics**
- Drive components in an electric vehicle
- Assembly and interconnection technology
- Module assembly and testing
- Thermal management
- Reliability



Source: Danfoss

Power electronics is the technology for the efficient conversion and control of electrical energy from the source to the consumer.

Power electronic applications range from a few watts to several gigawatts. In addition to the automotive and energy sectors, applications can be found in industrial and lighting technology, as well as consumer electronics.



Transportation technologies

- Trains
- Automotive
- Forklifts
- ...



Energy Technology

- Solar power plants
- Wind power station
- ...



Conveyor Technology

- Elevators
- Escalators
- ...

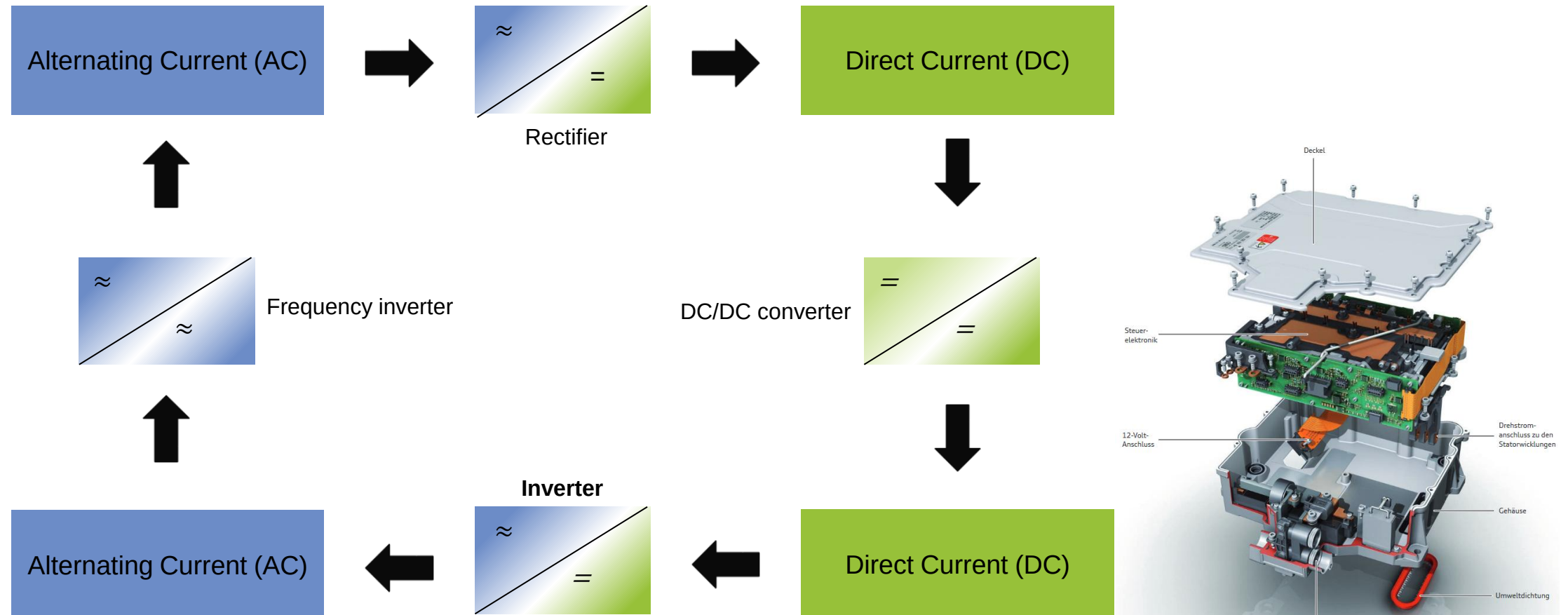


Other Fields

- Lighting technology
- Consumer electronics
- Welding technology
- Production technology

Power electronics convert electrical energy into the form required by the consumer.

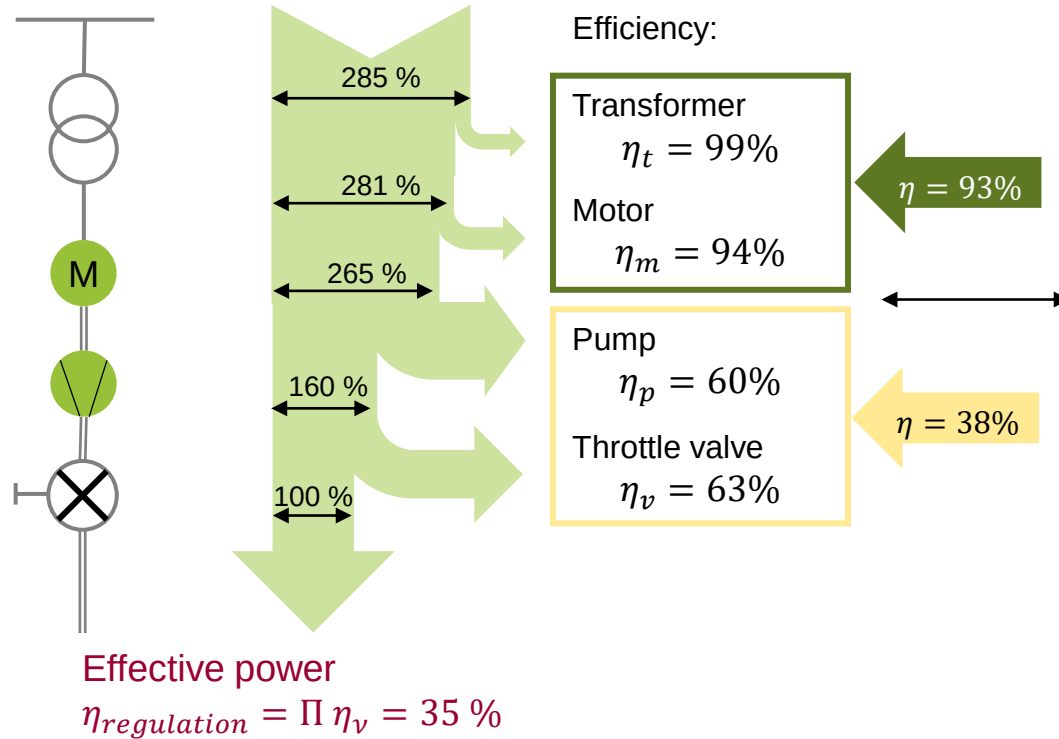
Power electronics have the task of controlling the flow of energy from the energy source to the consumer efficiently, reliably, need-oriented, and timed. For the control electronic switches are used, which are optimized for minimal conversion loss.



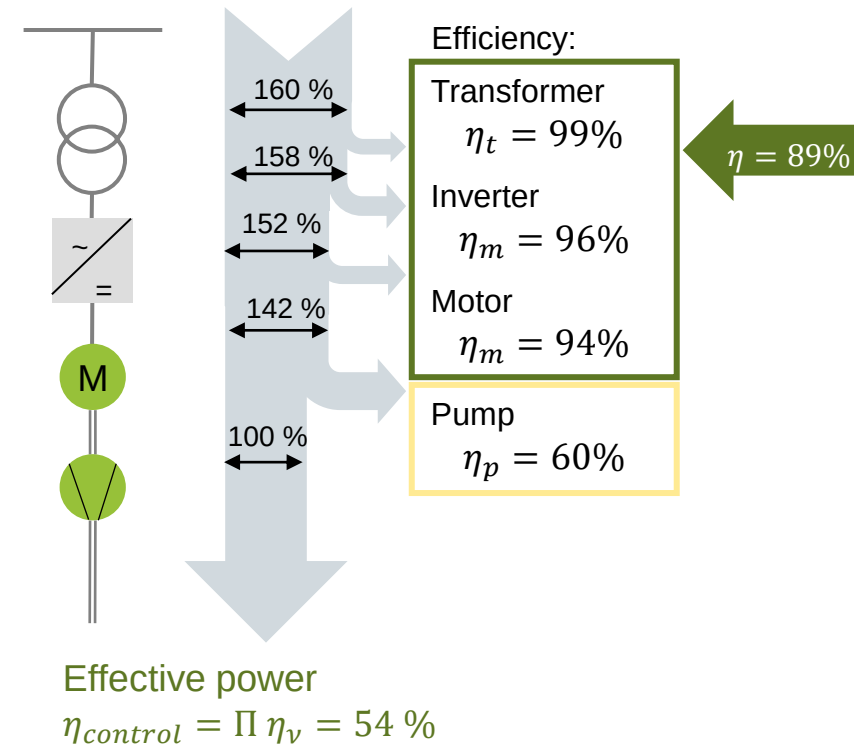
Source: Audi

The use of power electronic systems offers considerable potential for the sustainable and resource-efficient use of energy.

Volume Control via Throttle (without power electronics)

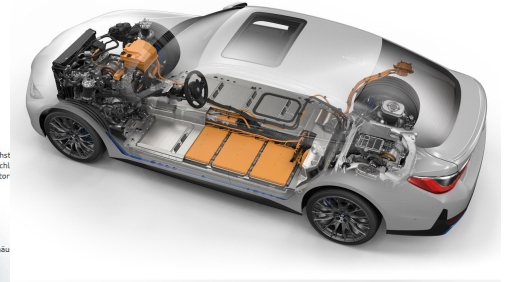
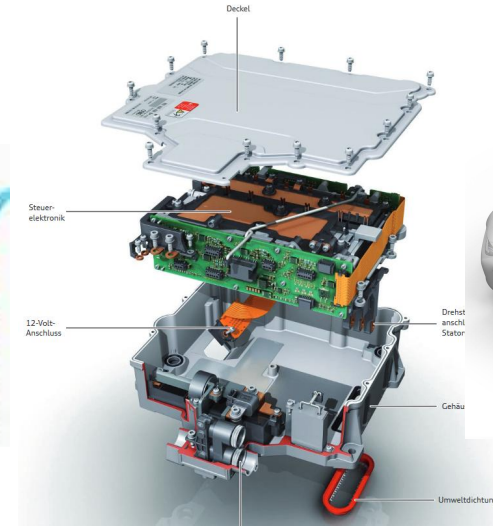


Volume Control via rotational speed (with power electronics)



➔ With the same power output, volume control results in a 44% decrease in power input or electricity costs ($(1 - \eta_{control} / \eta_{regulation}) = 44\%$), even though pure speed control has a lower efficiency ($\eta_0 = 89\%$) compared to power supply and motor efficiency ($\eta_0 = 93\%$).

The value creation system in power electronics comprises a multitude of manufacturers and competencies - from component manufacturers to system integrators.



Components

- Bare Dies (Semiconductor Chips)
- Inductivity
- Capacity
- Circuit carrier (DCB)

Power Module Packaging

- Bonding
- Die attach
- Encapsulation
- Planar assembly

Assembly Group

- Rectifier
- DC/DC converter
- Inverter
- Frequency inverter

Subsystem

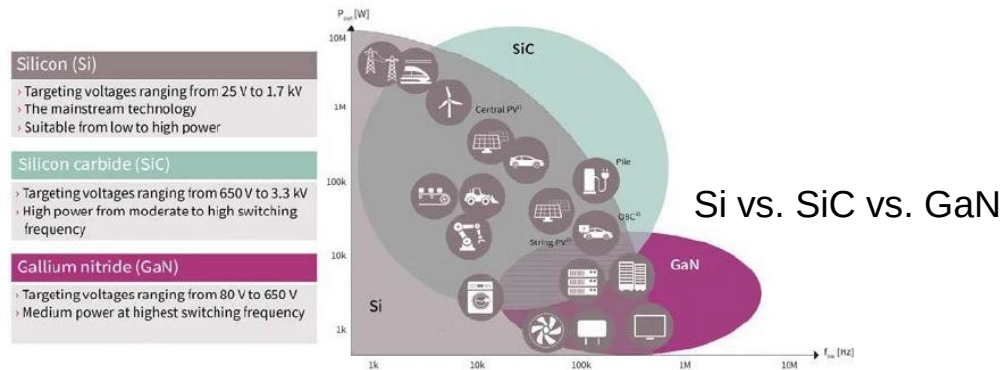
- Drive unit
- Power supply
- Solar inverter

System Integration

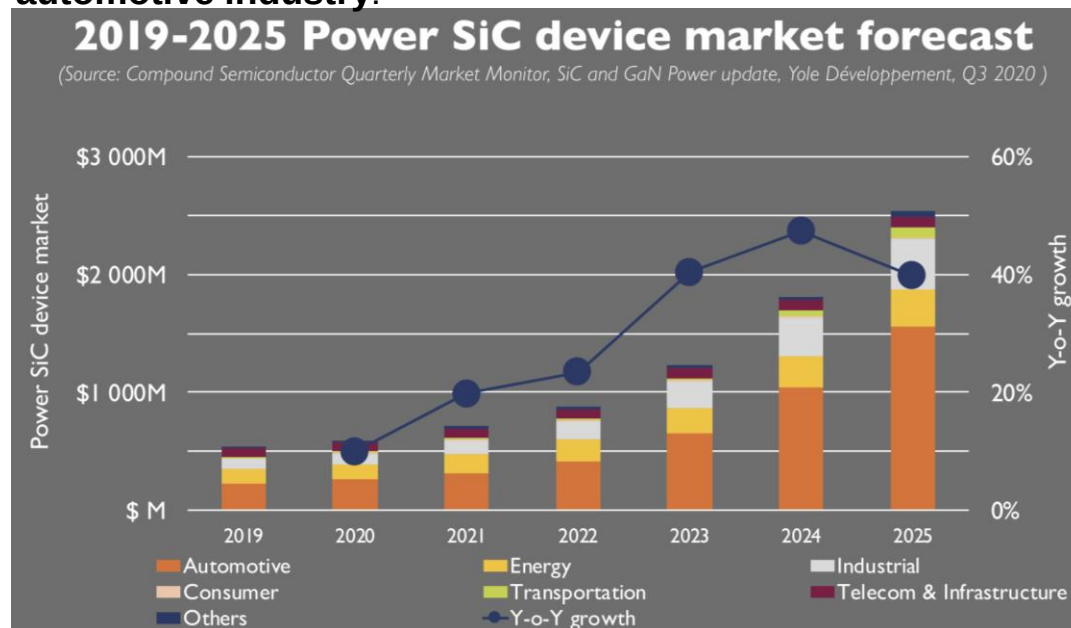
- Automotive
- Healthcare
- Consumer electronics
- Machining tools
- Construction technologies

Source: Rogers, Fraunhofer ISIT, Audi, Danfoss, Infineon, BMW

The application of new semiconductor materials (SiC) reduces energy consumption in main inverter by 69% or increases power while maintaining the same footprint.



The semiconductor market is exponentially driven by the **automotive industry**:



In order to **enable SiC chips** with high junction temperatures and high temperature differences several main technologies need to evolve:

■ New materials, designs and processes for:

Substrate

High thermal conductivity ceramics (AlN , Si_3N_4)

Encapsulation

High-temperature epoxy or silicon gel

Die attach

Silver or copper sintering

Interconnections

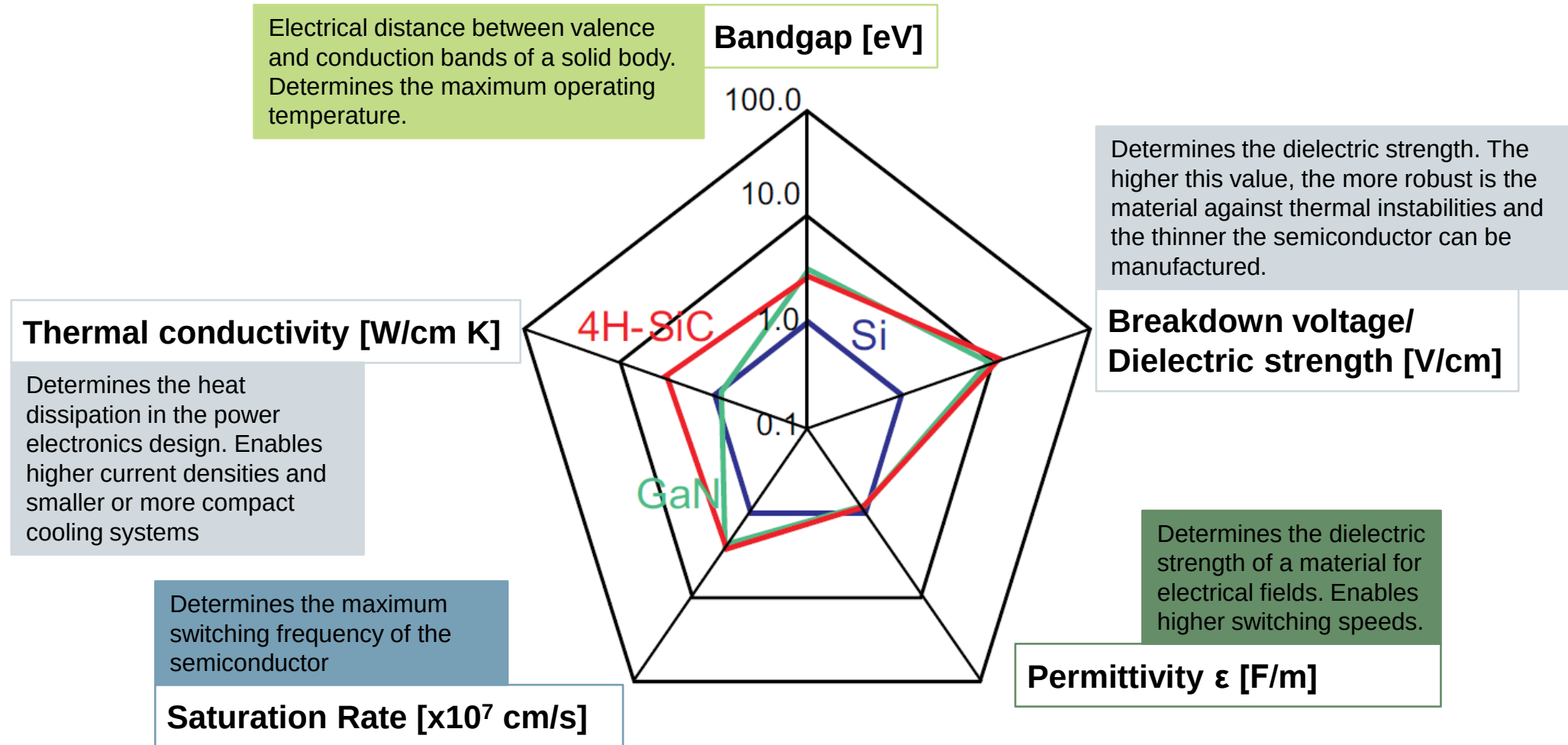
Low-inductance and high reliability interconnections

Further trends are:

- Functional integration
- Double-sided cooling
- Overmolded packages
- IMS (Insulated metal substrate)
- PCB embedding

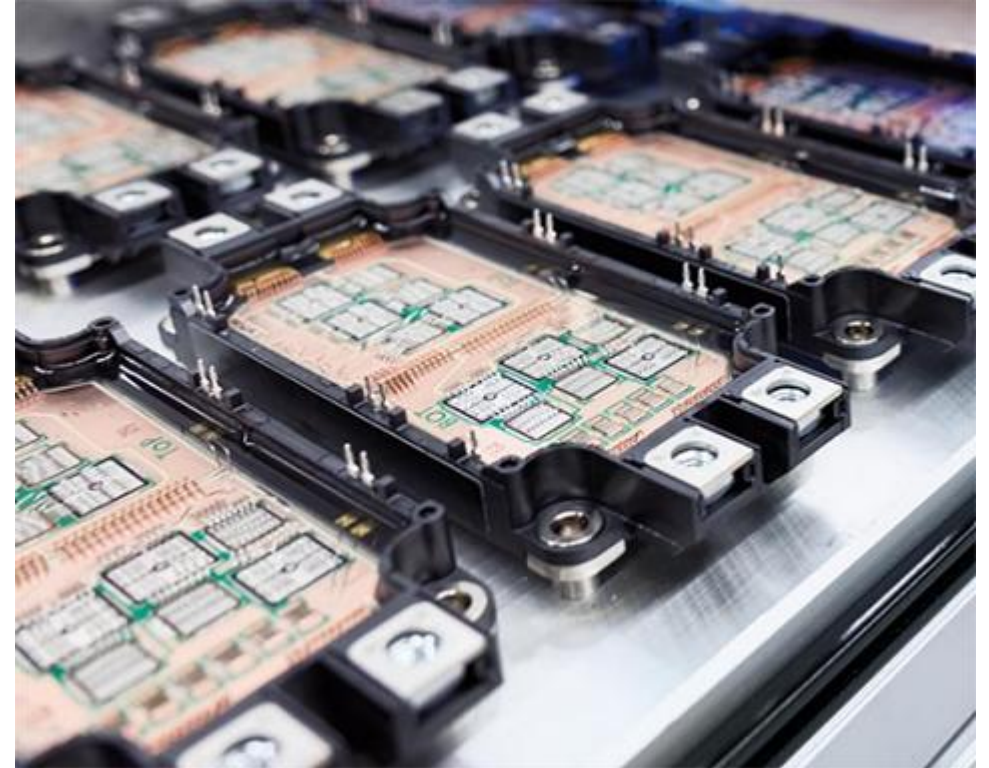
Source: Infineon, Yole

New materials expand the performance spectrum of power semiconductors due to improved physical properties.



Power electronics must be adapted to the electric drives just like the control and cooling systems.

- Introduction to power electronics
- **Drive components in an electric vehicle**
- Assembly and interconnection technology
- Module assembly and testing
- Thermal management
- Reliability

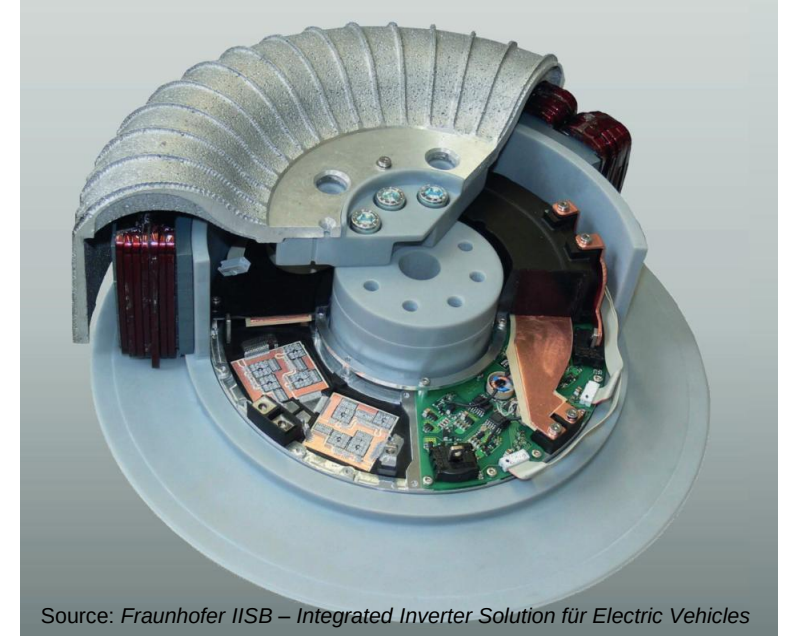


Source: Danfoss

Currently, the production of power electronics components is characterized by a high proportion of manual manufacturing steps.

Power electronics in electric vehicles

- Fundamentals and structures comparable to those used in stationary power electronics such as inverters and power units
- Especially for electromobility with a focus on high-integrated and lightweight power electronics components
- High demands on reliability (shock, temperature changes)



Manufacturing of power electronics for electric cars:

- High proportion of manual processes
- Mass production in automotive scale required
- Higher quality through better and automated processes



Electronic development for the electric vehicle encompasses not only the electric motor and energy storage, but also the control and drive components.

Power Electronics

- Inverter
- On-board power converter
- Charger
- Vehicle control

Potentials

- Efficiency
- Lower volume
- Reduced costs

Electric Drive

- PSM, ASM
- Central motor
- Wheel hub motor
- Integrated motor

Potentials

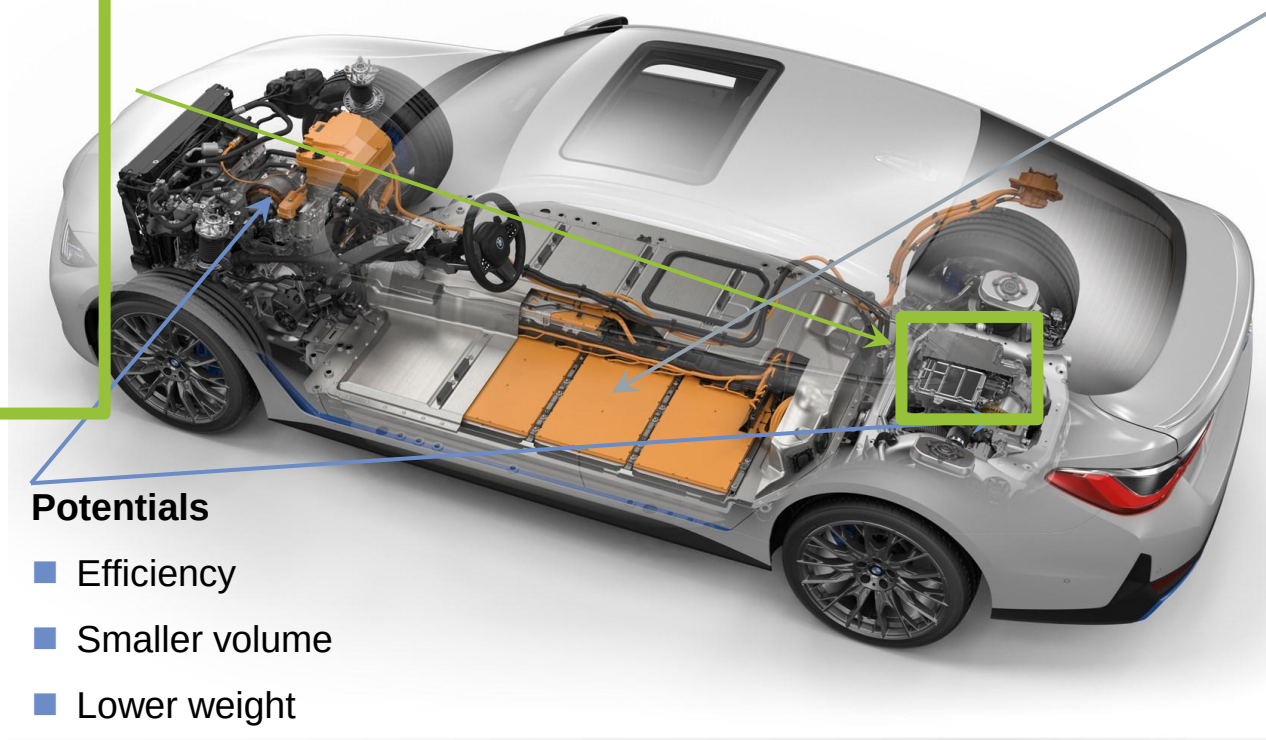
- Efficiency
- Smaller volume
- Lower weight
- Reduced costs

Energy Storage

- NiMH, Li-Ion, SCAP
- Cooling
- Battery management

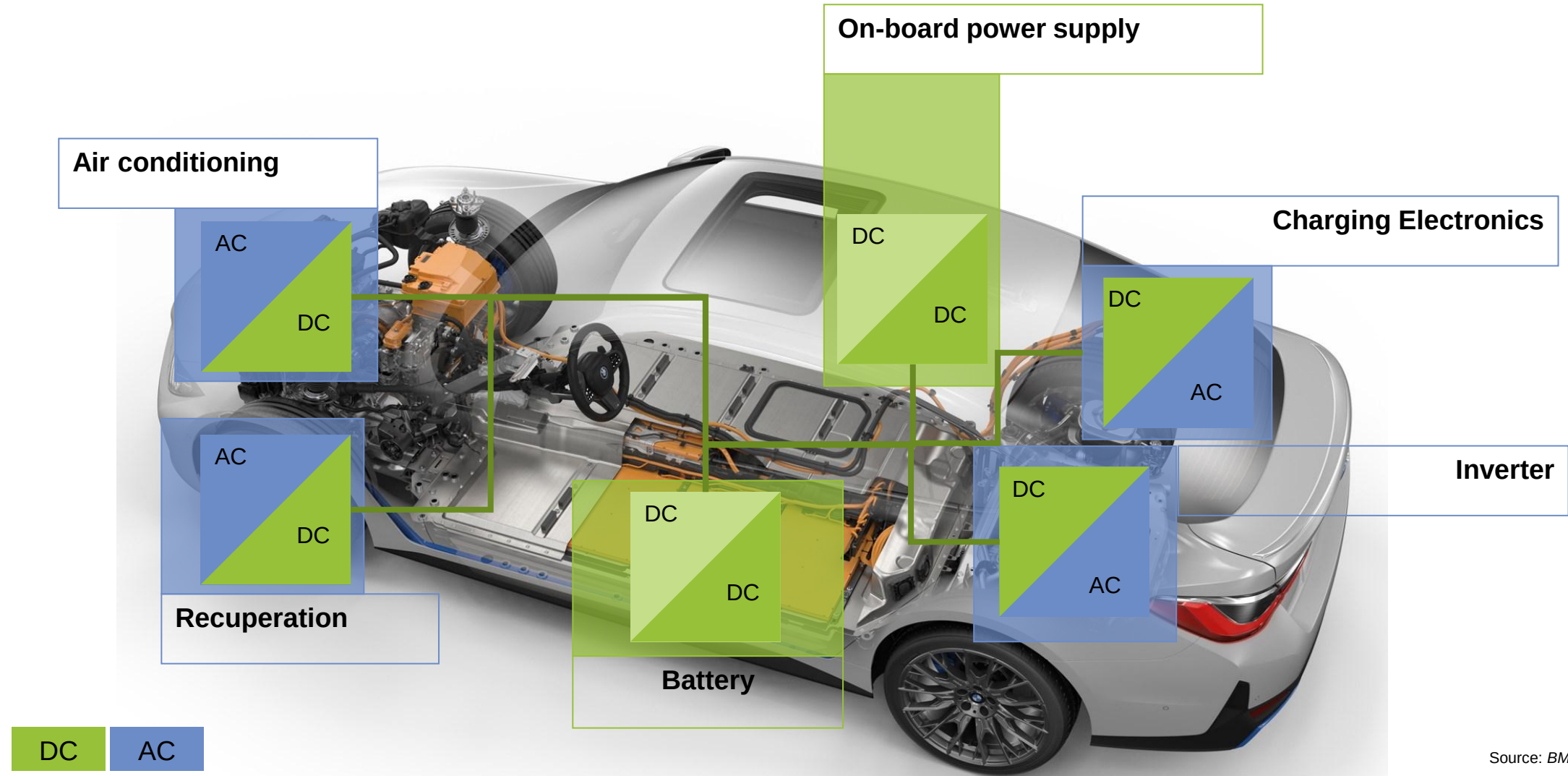
Potentials

- Higher energy density
- Affordable materials
- Lower weight



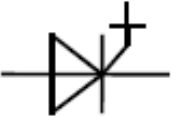
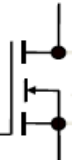
Source: BMW i4

From the perspective of power electronics, this particularly concerns the inverter and DC-DC converter for the electric drive and charging electronics.

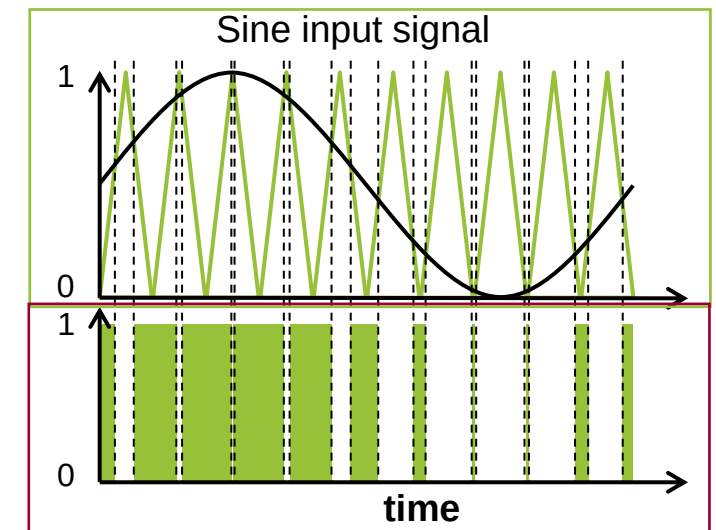
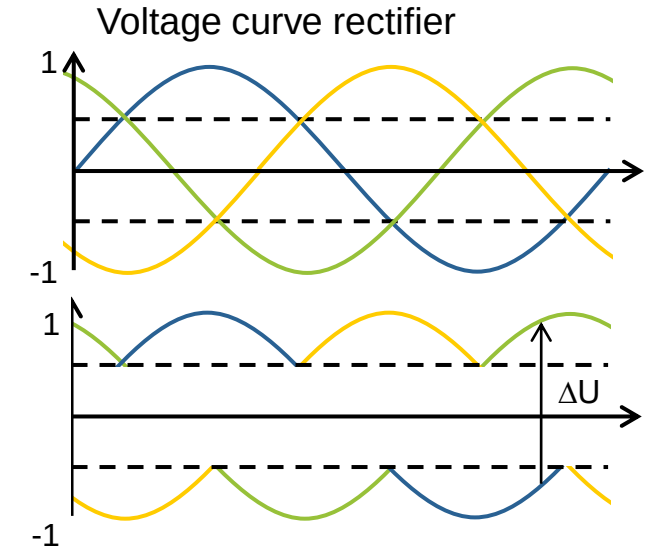
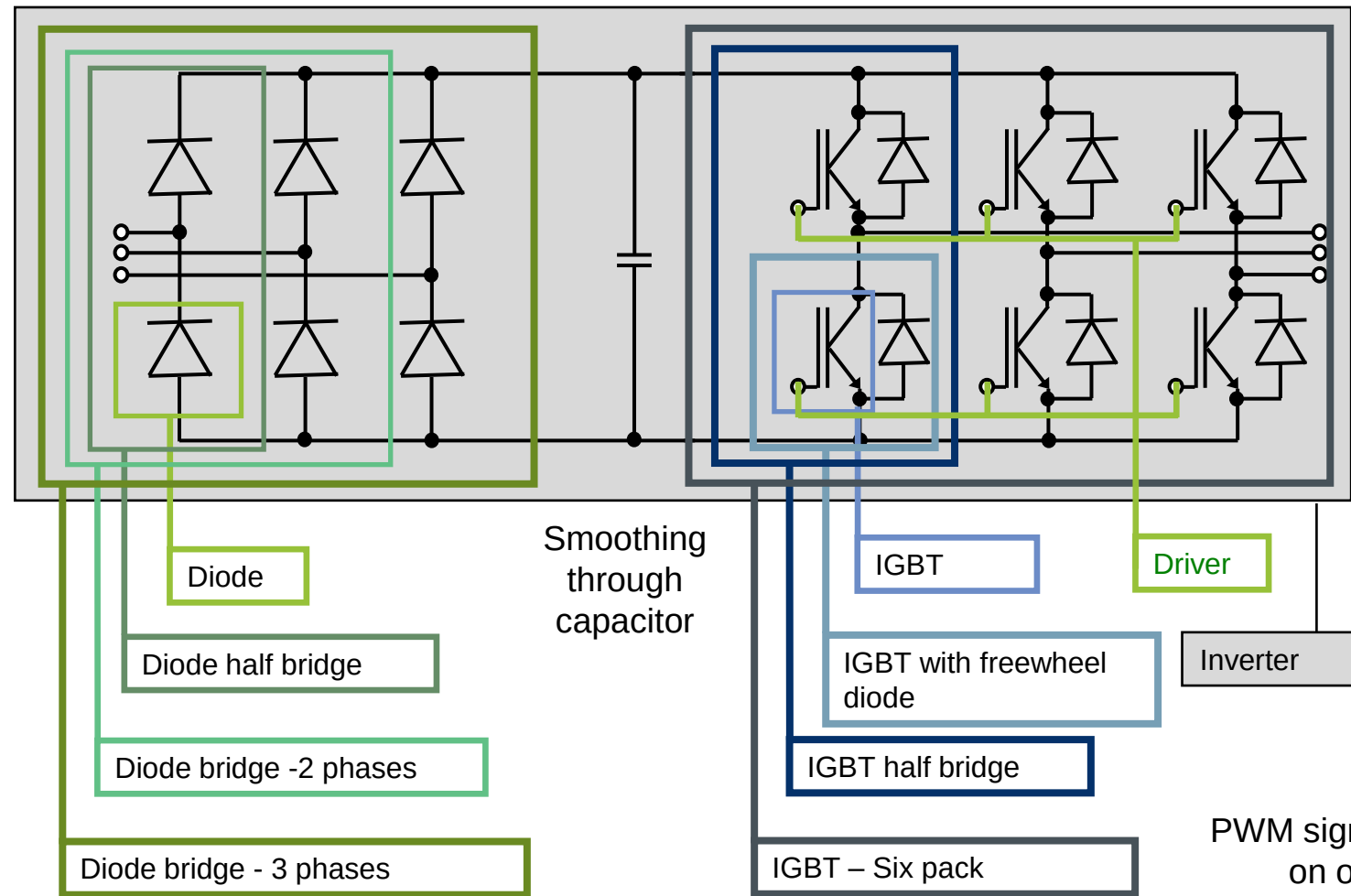


Source: BMW i4

The presentations in this lecture concentrate on the most commonly used power electronic components at present.

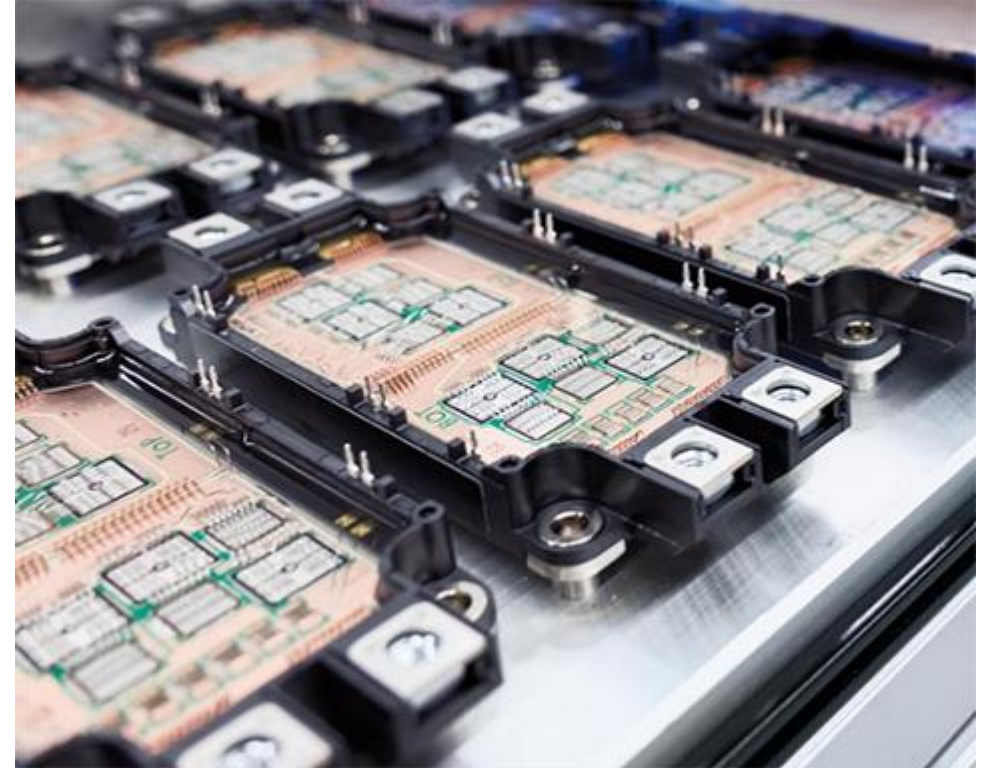
Type	Switching capacity	Frequency	Application	Circuit symbol
Thyristor	Very high	Low	High-power switch	
GTO (Gate Turn Off)	High	Medium	High-power switch with lower cycle times	
IGBT (Insulated Gate Transistor)	High	High	Switching device, inverters	
MOSFET (Metal Oxide Field Effect Transistor)	High	Very high	Most popular transistor for ICs	
Diode	Very low – very high	Very low – very high	AC-DC conversion	

Using a frequency converter, which consists of a rectifier and an inverter, the electrical switching functions of power modules can be demonstrated.



Power electronics must be adapted to the electric drives just like the control and cooling systems.

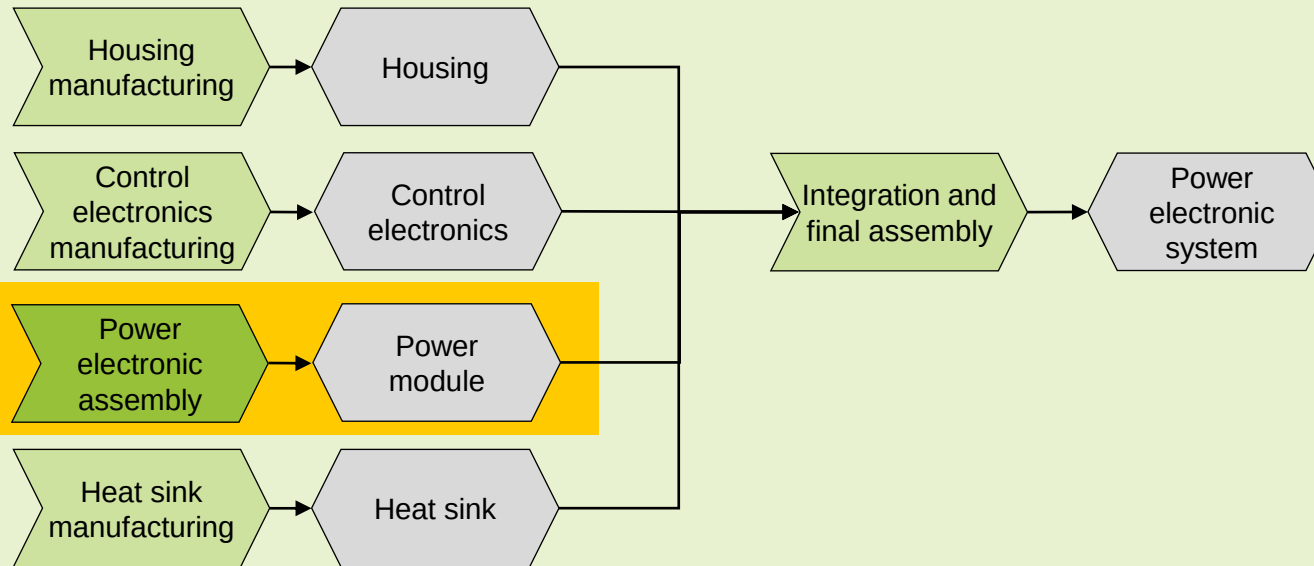
- Introduction to power electronics
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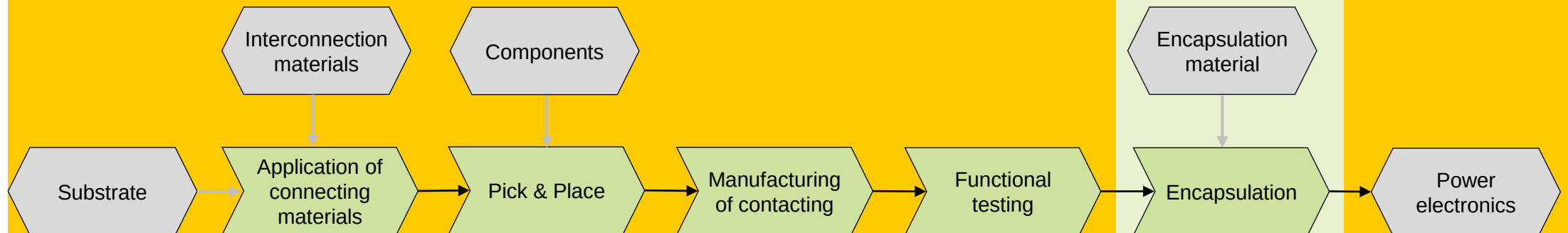
Source: Danfoss

The CONSENS Model conveys an overview of the manufacturing chain and the classification of power electronics into an overall system, as it is addressed in the lecture.

Level 1



Level 2



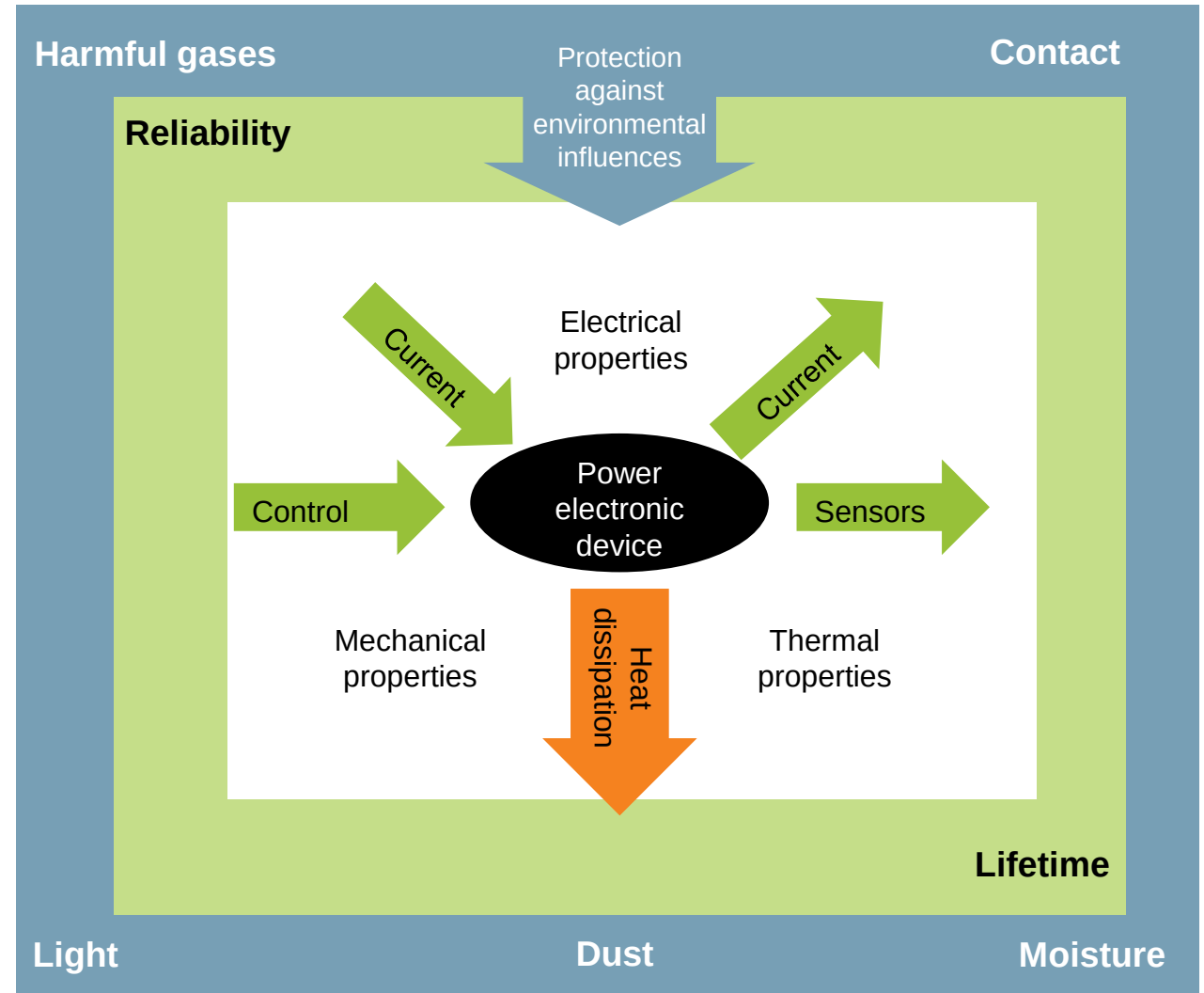
The assembly and connection technology fulfils fundamental tasks for the realization of reliable power modules.

Fundamental functions of all designs

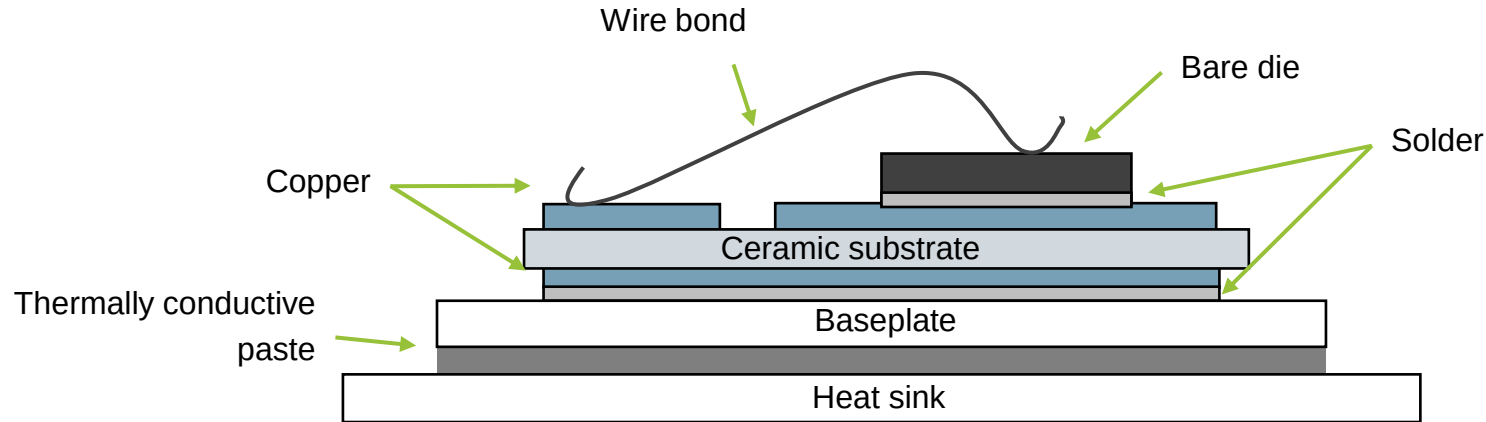
- Heat dissipation
- Connections for current flow
- Functional insulation
- Mechanical protection
- Protection against environmental influences (e.g., moisture, shock, and vibration)
- Mounting device
- Ensuring reliability and lifetime

Additional functions of many designs

- Basic insulation
- Connections for control contacts
- Sensors for protection



The classic structure of power modules fundamentally differs from the construction of conventional flat assembly groups.



Standard die attach technologies

- Soldering
- Wire Bonding
- Pressure contact

Alternative technologies for die attach

- Sintering
- Conductive gluing
- Trans liquid phase soldering
- Spring contact
- Direct bonding

	Heat sink		Baseplate	Substrate			Bare Die		
Task	Heat dissipation		Heat spreading	Electrical isolation			Function realization		
Material	Al	Cu		Al ₂ O ₃	AlN	Si ₃ N ₄	Si	SiC	GaN
CTE [10 ⁻⁶ /K]	23,1	16,5		6,8	4,7	2,7	2,6	4,3	5,6
Thermal Conductivity [W/mK]	235	400		19-30			150		

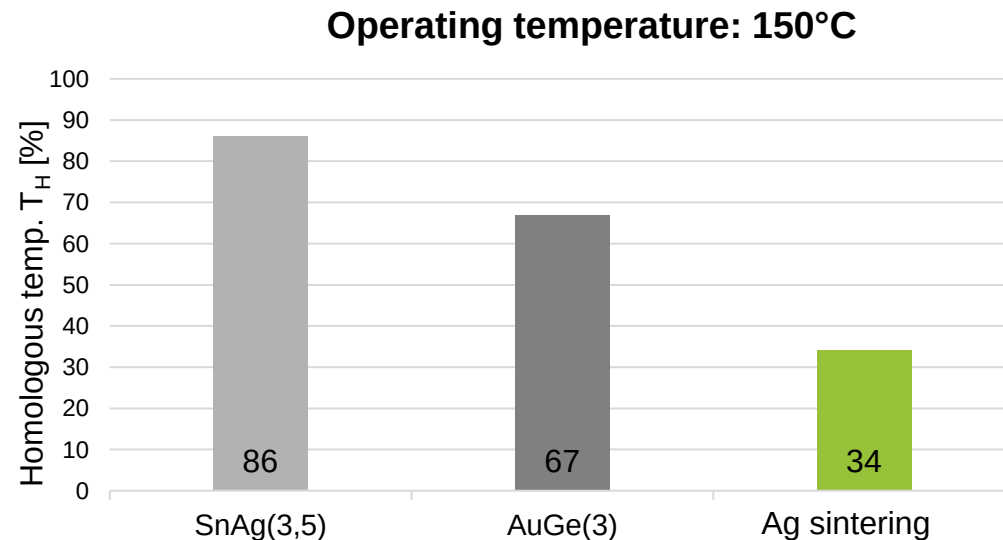
Excursus: The concept of homologous temperature enables an assessment of the reliability of a joint connection under thermo-mechanical stress.

The homologous temperature T_H is defined as the ratio of operating temperature to melting temperature in [K]

$$T_H[\%] = \frac{T_{\text{operating temperature}}}{T_{\text{melting point}}} \cdot 100\%$$

A **low homologous temperature** indicates a high reliability::

- < 40%: mechanically stable
- 40-60%: creeping area, sensitive to stress
- > 60%: mechanical stresses cannot be permanently endured and lead to the destruction of the connection



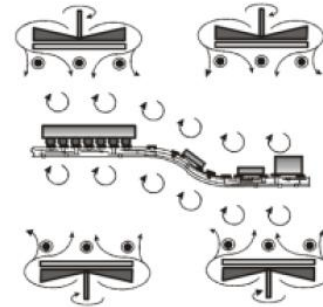
	SnAg (3,5)	Ag Sintering
Melting point [°C]	221	961
Melting point [K]	494	1234
Operating temperature [°C]	150	150
Operating temperature [K]	423	423
Homologous temperature [%]	86	34
Thermal conductivity [W/mK]	70	240
Electrical conductivity [MS/m]	8	41
Layer thickness [μm]	~90	~20
CTE [ppm/K]	28	19
Tensile strength [MPa]	30	55

For the die-attach in power electronics, convection soldering and condensation soldering are the two most important soldering processes.

Reflow soldering

1. Solder paste application
2. Component placement
3. Melting of the solder during soldering process with:

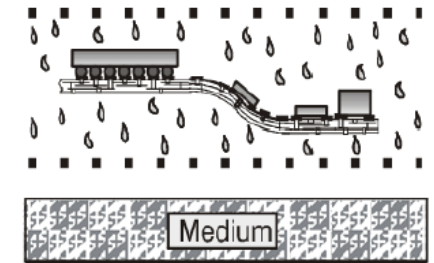
Convection soldering



- The airflow is heated above the heating system and defined by nozzles (slot nozzles) onto the soldering assembly that has been applied with solder paste.
- Through even shaping, shadowing and shading effects are largely avoided.
- The processing of complex assemblies with components, whose connection structures are hidden under the component, is possible.
- Only a very small ΔT in the peak zone.

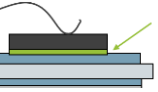
or

Condensation (vapor-phase) soldering



- Latent heat from condensing vapor heats the solder up to the boiling temperature of the fluid.
- Heat input independent of surface properties of the soldering fluid.
- The fluids used minimize oxidation as they act in a protective gas atmosphere.
- Overheating of components is excluded, as fluids do not exceed their boiling point under ambient pressure.
- Soldering temperature is dependent on the choice of fluid.

Convection soldering and vapor-phase soldering have different advantages.



Convection soldering



or

Condensation (vapor phase) soldering



**There are several advantages
for each process:**

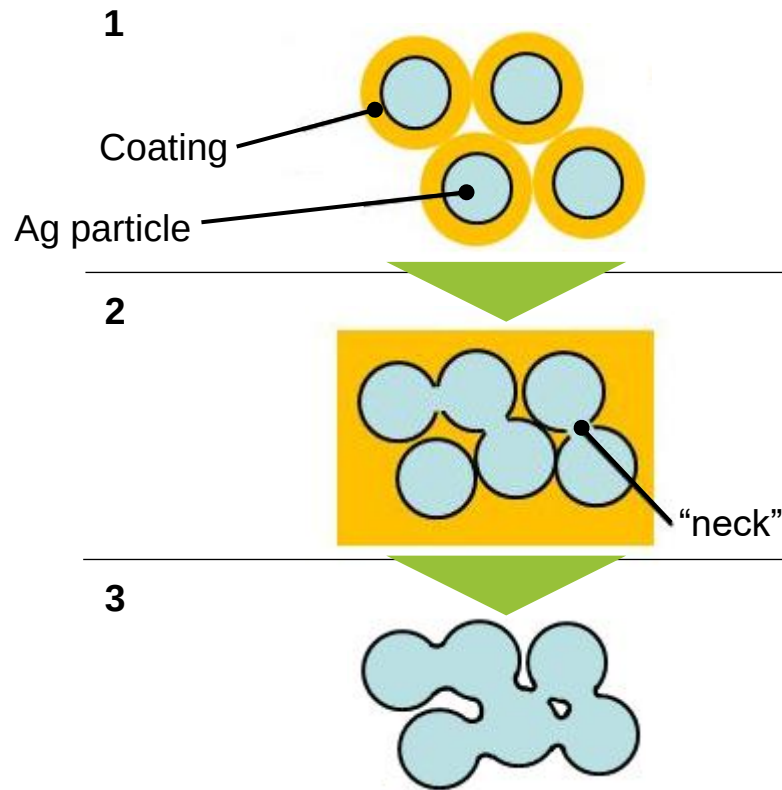


- Uniform temperature distribution across different circuit carriers
- Smaller ΔT compared to infrared soldering in the peak zone
- Reduced shadowing effects
- Lower risk of thermal overload on assemblies

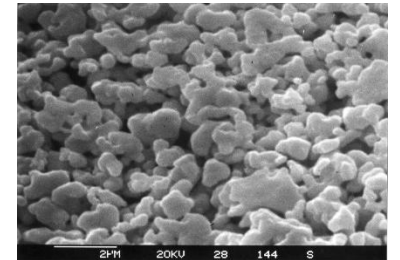
- Homogeneous temperature distribution (nearly ΔT in the peak zone)
- High reproducibility
- Simple profiling
- No risk of local thermal overload due to preset working temperature
- Soldering of large thermal masses (e.g., heat sinks)

Sintering is the creation of a dense body by pressing powder and eliminating void space.

Sintering: The sintering technique combines two fine-grained ceramic or metallic materials, usually under high pressure and at temperatures below the melting point of both materials. The surfaces must therefore be coated with an inorganic coating so that the particles do not melt together.

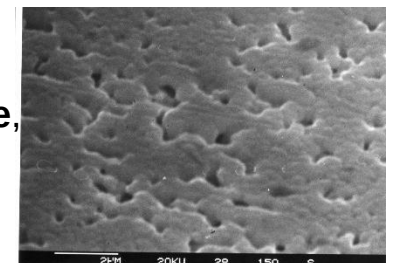


When the silver powder is heated, the wax starts to melt, allowing the particles to come into contact with each other. This occurs at **about 120°C**. Solvents in the paste also evaporate.



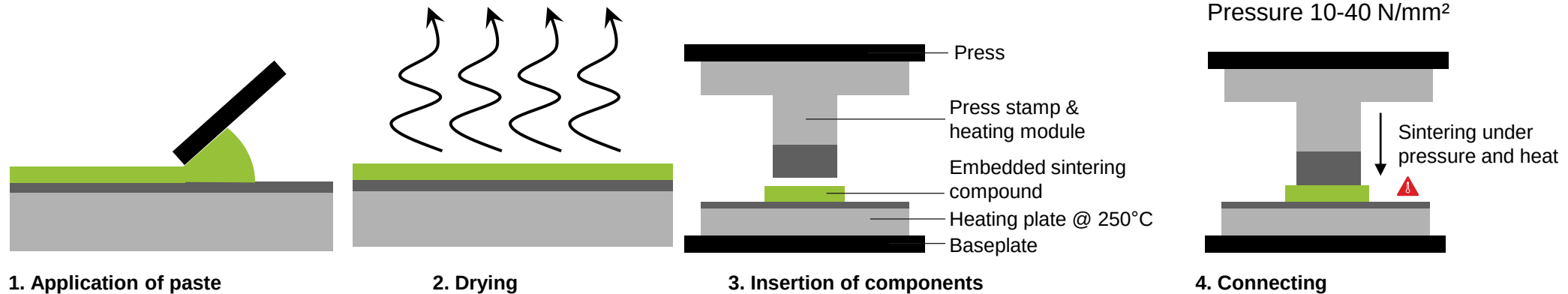
At higher temperatures of **over 200°C**, the wax evaporates and pressure (normally around 10-30 MPa) promotes the growth of sintering between the silver particles.

Some **pores remain** in the sinter structure; the porosity can be precisely controlled through process parameters such as **pressure, temperature, and time**.



Sintered connections enable the rising reliability requirements of high-performance modules to be met in terms of thermal and electrical robustness.

Process chain of silver sintering:



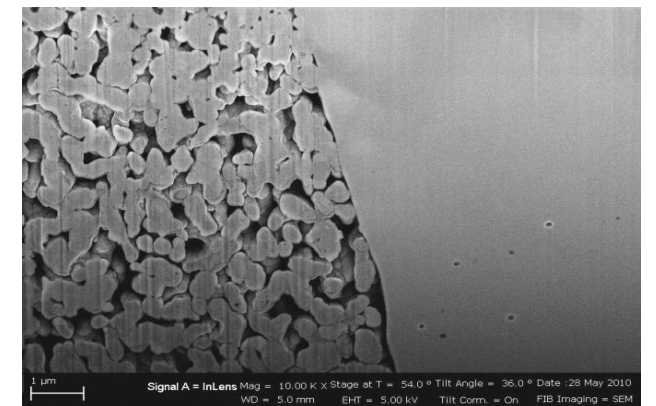
Advantages

The material properties of a sinter layer are superior to a soldered joint in all aspects:

- Significantly lower homologous temperature T_H (therefore higher stability under thermo-mechanical alternating stress)
- Higher electrical conductivity
- Higher thermal conductivity
- Greater mechanical strength

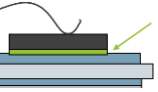
Disadvantages

- Reparability/Re-workability of sintered assembly groups in general is not possible
- Material costs are relatively high
- Costly tooling
- Stress-input during process



SEM image of a sinterable silver material after 1200 cycles.

Diffusion soldering refers to the realization of high-temperature chip connections through the formation of intermetallic phases.

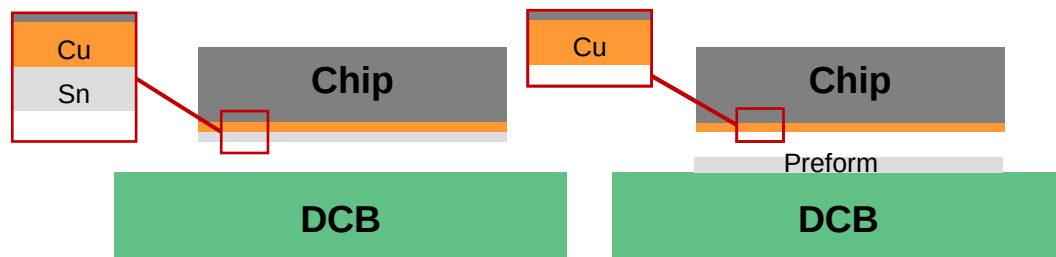


Diffusion soldering: When heated to the melting temperature of a material with a low melting point (liquid phase), a combination with a material with a high melting point leads to diffusion and a reaction of both materials in the form of the formation of intermetallic phases (IMP) with a higher melting point than the low melting point material (e.g., Transient Liquid Phase Bonding (TLPB) or Transient Liquid Phase Soldering (TLPS))

- Isothermal solidification of the liquid phase in the material with the lower melting point through diffusion
- Diffusion from the material with the lower melting point into the material with the higher melting point and vice versa
- Very thin connection layers ($5\mu\text{m} - 50\mu\text{m}$)

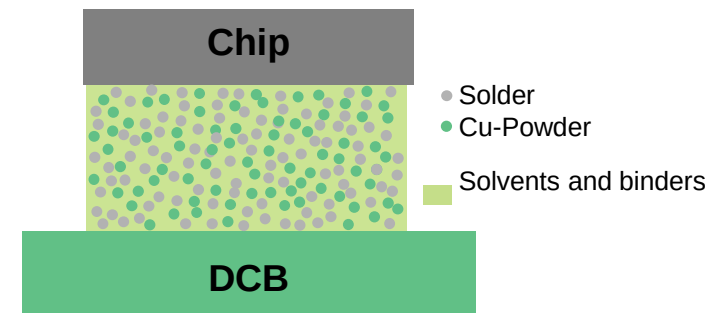
TLPB – Trans Liquid Phase Bonding

- Metal-coated (e.g., tin) chips and substrates
- Layer build-up typically with thin-film deposition techniques
- Use of preforms



TLPS – Trans Liquid Phase Soldering

Use of solder paste for wetting, possibly mixed with high-melting metal particles in the paste printing process



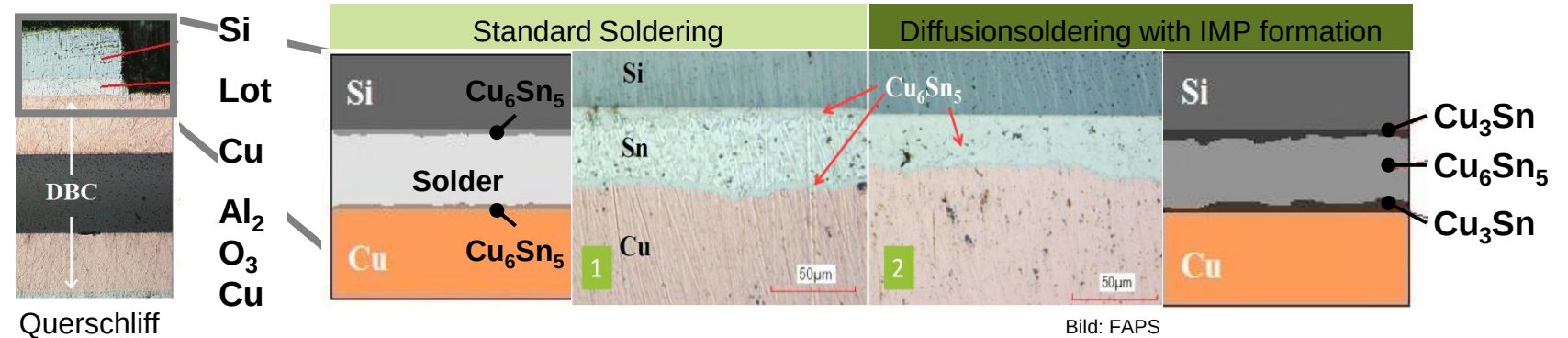
Different material combinations, which differ in terms of process management, costs, and temperature stability, enable the realization of diffusion soldering connections.

Material	Melting point [°C]
Ag	961
Au	1064
Cu	1083
In	156
Sn	232
Cu_6Sn_5	415
Cu_3Sn	676
Au_5Sn	519
AuIn^*	>600
$\text{Ag}_3\text{Sn}^{**}$	480
$\text{Ag}_{85}\text{Sn}_{15}$	650

*only under inert or reducing atmosphere

**intermetallic for high-temperature chip connections

Example: Formation of Cu_6Sn_5 - and Cu_3Sn -phases



Advantages

- Higher melting point
- Lower homologous temperature T_h than soldering connections
- High reliability and long lifespan

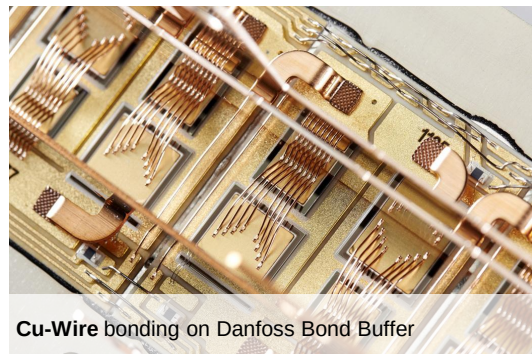
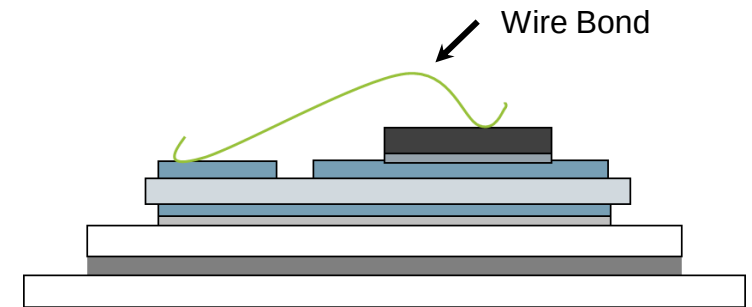
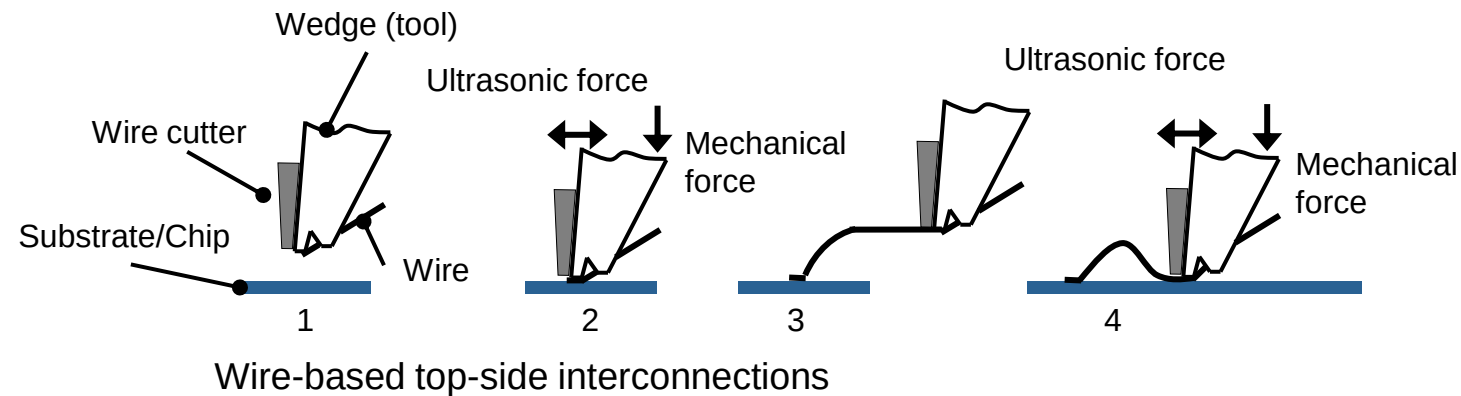
Disadvantages

- Increased joint bonding
- Complex process optimization
- Brittle intermetallic phases
- Only TLPS: High demands on the material quality of the joining partners
- Long process time

In US-wire bonding, similar or dissimilar materials can be joined through the application of pressure and ultrasound.

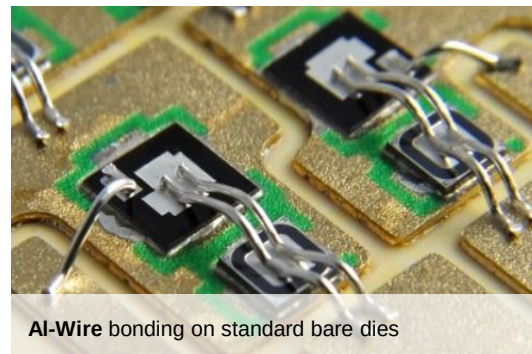
The dominant combination in power electronics is the **wedge/wedge-ultrasonic bonding** with thick aluminum wires (wire thickness > 100 μm). Increasingly, flat wires as well as copper wire bonds are being used.

Process sequence



Cu-Wire bonding on Danfoss Bond Buffer

Source: Danfoss



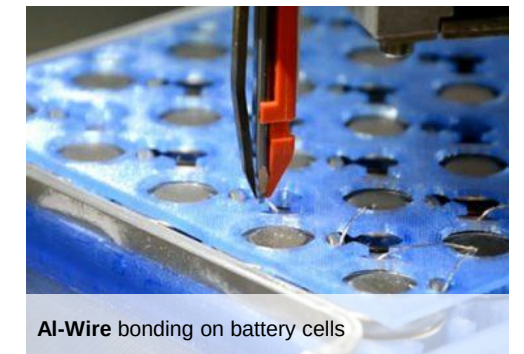
Al-Wire bonding on standard bare dies

Source: Semikron



Al-Ribbon bonding on standard bare dies

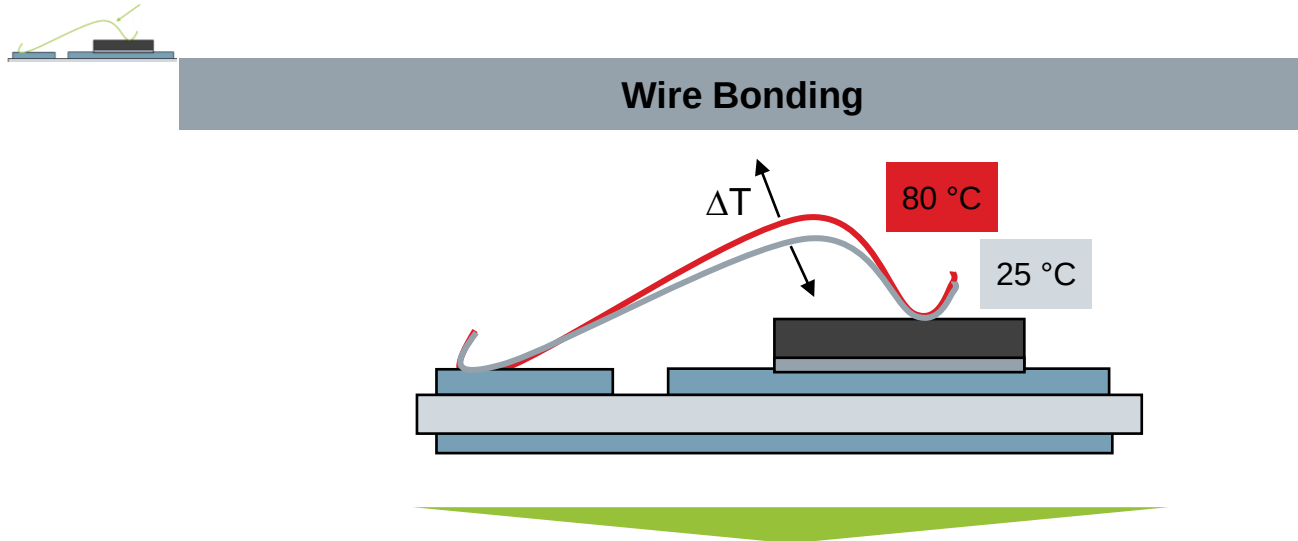
Source: F&S Bondtec



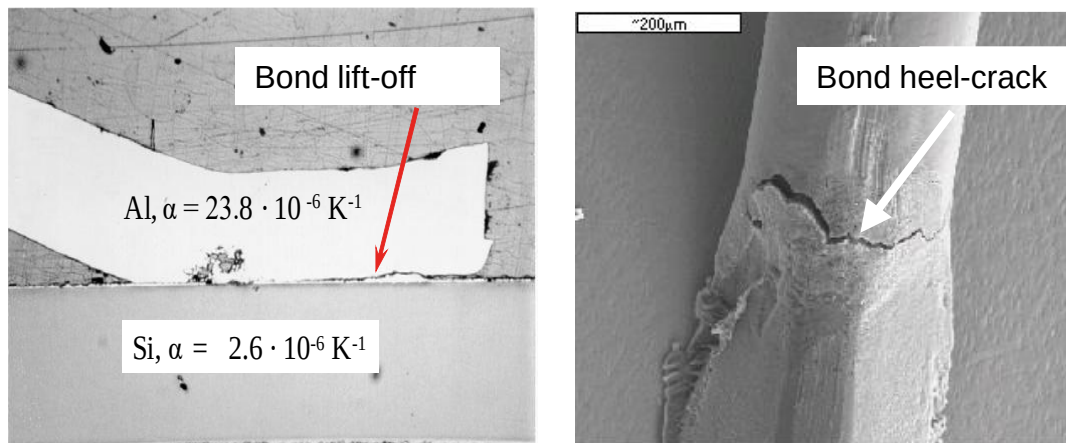
Al-Wire bonding on battery cells

Source: F&S Bondtec

Wire bonding is a universal method, even when dealing with sequential processes for overcoming chip contact and wiring.



Typical failure modes of bond connections



Source: ECPE-Cluster „Failure analysis in microelectronic systems“; 2012

Advantages

- Flexible connection in terms of housing, substrate, and chip geometry
- Contacting of temperature-sensitive components and materials
- High reliability, especially for Al/Al or Cu/Cu contacts
- Repairability of the connection
- Cost-effective wire material
- Connection of different materials possible

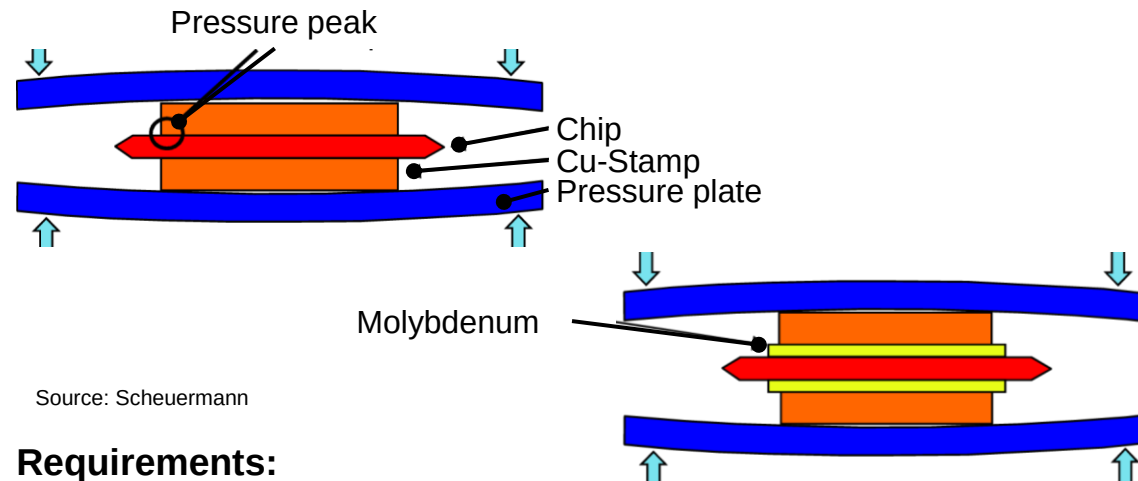
Disadvantages

- Sequential process
- Mechanical damage, surface cleaning, and roughness can hinder connection formation
- Secure processability of the connection not always possible

In pressure contact, a chip is pressed between two contact stamps with defined pressure and is thereby electrically and thermally contacted.

Pressur contacting

Unlike soldering, sintering, and wire bonding, pressure contact does not involve a molten connection, but rather a force-fit connection. The connection is made through pressure without the need for elevated temperatures, as with solder or wire connections. This explains the high reliability and long lifespan of pressure contact elements.



Source: Scheuermann

Requirements:

- Uniform distribution of pressure on the chip – avoidance of local pressure points
- Contact stamp impression smaller than the chip surface requires a certain distance from the edge

Advantages

- Optimal thermal connection through bilateral cooling possible (heat sink level)
- High reliability and long lifespan due to reduced thermomechanical stress

Disadvantages

- High demands on the material quality concerning planarity in the pressure area
- Lack of heat dissipation
- Additional manufacturing costs for the pressure contact setup per chip

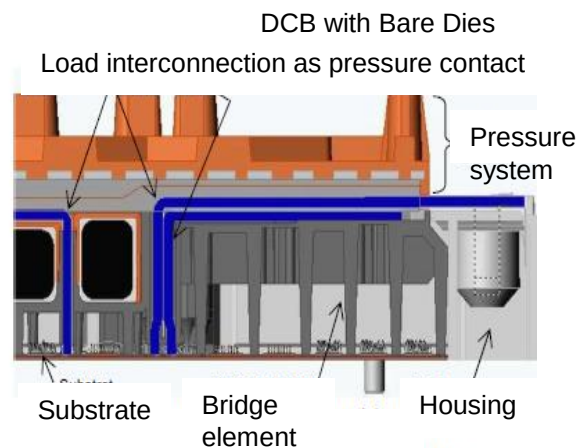
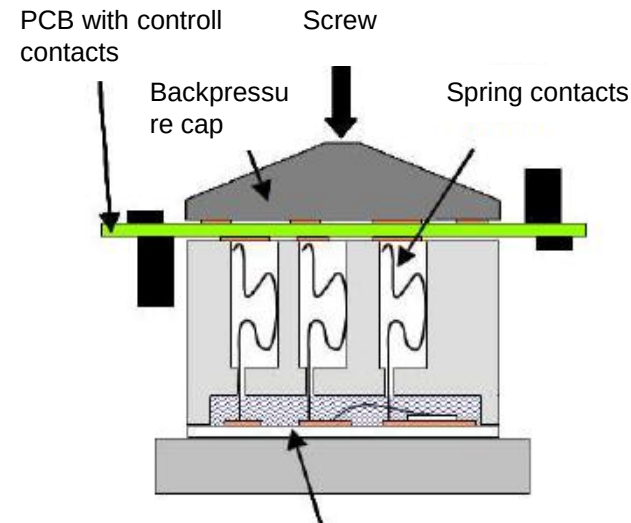
Spring contacts represent an alternative for the construction of load, control, and sensor contacts, which establish the connection to the superordinate overall system.

Spring contacting

Spring contacts are made from elastically deformable metal sheets or foils, which are placed in a fixed position in the housing. They establish an electrical connection via metal spring force between the power electronics subassembly and the control electronics circuit board or peripheral devices.



Source: Scheuermann Load interconnections with spring contacts



Advantages

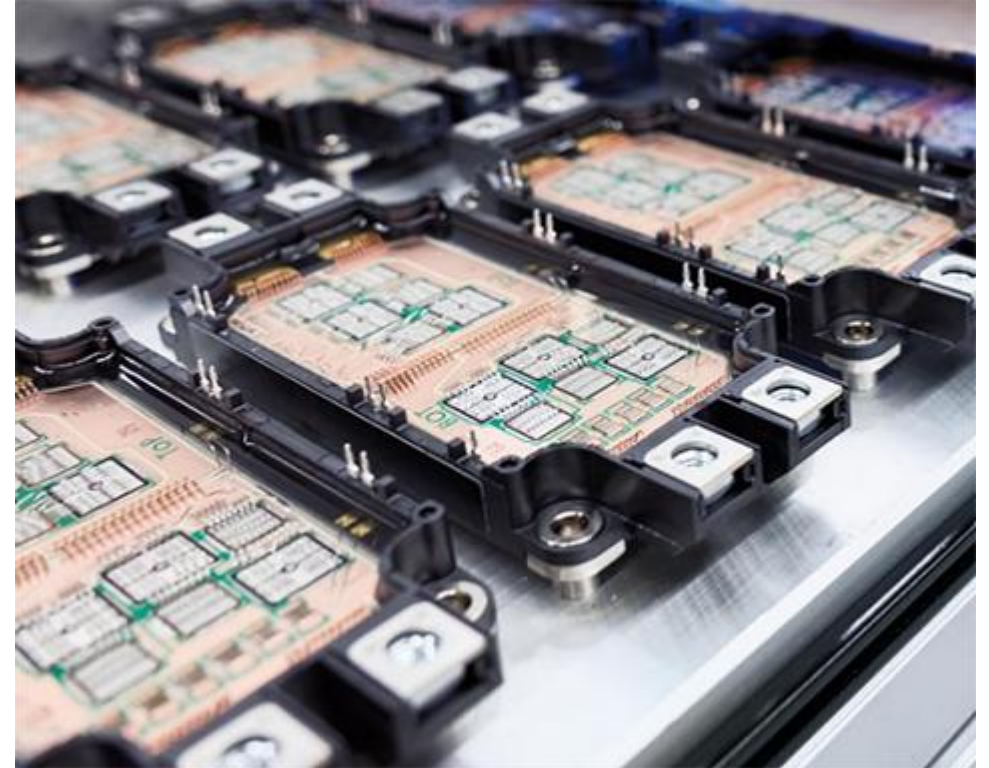
- Simple assembly on circuit boards and control electronics
- No thermal load fatigue due to vibration is possible
- Greater tolerances in the mechanical overall assembly can be compensated

Disadvantages

- Lower current carrying capacity per area compared to molten connections
- Contact area is not achievable with sufficient pressure

Power electronics must be adapted to the electric drives just like the control and cooling systems.

- Introduction to power electronics
- Drive components in an electric vehicle
- Assembly and interconnection technology
- **Module assembly and testing**
- Thermal management
- Reliability



Source: Danfoss

The variety of different forms for power components can be sorted based on current and voltage.

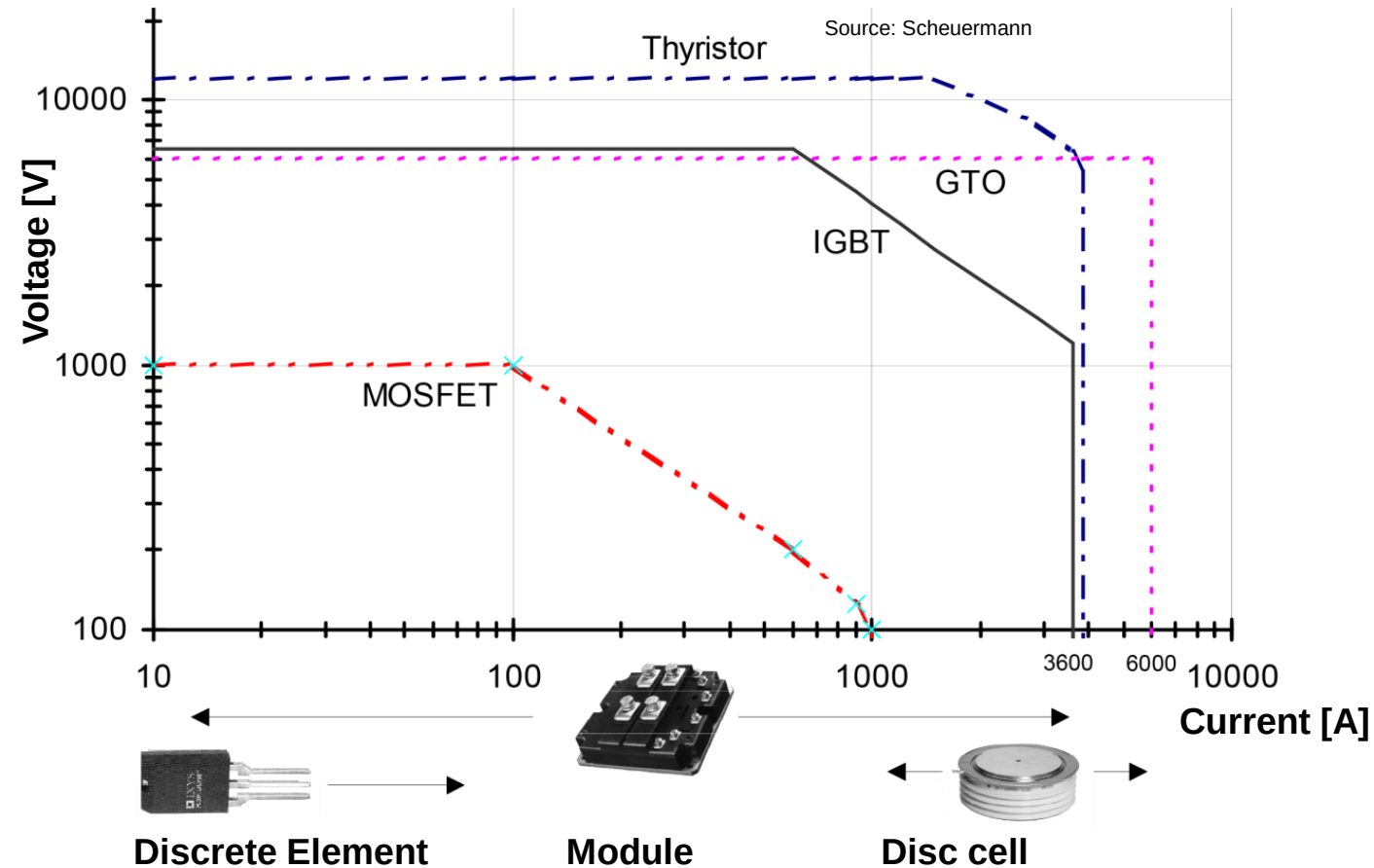


Bild: Semikron

The range of power electronics covers a wide performance spectrum, from a few watts up to the megawatt range. The currents reach from 1 A to 6000 A and more, and the voltages are between 80V and 10.5 kV.

The transistor outline (TO) design and its variants dominate the discrete designs in the low power range.

TO-Package

In the case of discrete component forms, the power chip is mounted on a 'leadframe' which carries all 'leads' (connection legs) over bridges and connects the modules into a band via further stages. After bonding, the construction of an artificial customer module is completed with the transfer molding process. Only at the end of the process are the bridges between the pins removed by punching.

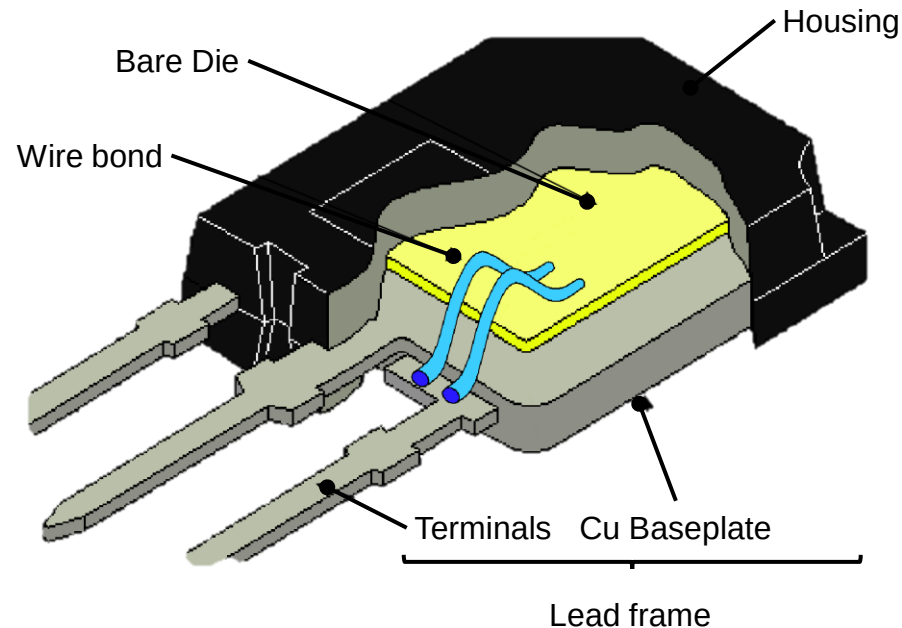


Image: Scheuermann

Advantages

- Simple internal structure
- Small size
- High production numbers (daily several million)
- Low production costs due to high standardization

Disadvantages

- Only suitable for small and medium power levels
- Lacking insulation of the components against the cooling surface
- Direct chip-on-heat-sink assembly leads to reduced lifespan

In disc cells, round thyristors, diodes, and GTOs are used for very high currents and/or voltages.

Disc cell

The basic principle of a disc cell is based on the pressure contact technique. A cathode-side one-sided pressure contact presses on the chip and thus makes the thermal and electrical contact.

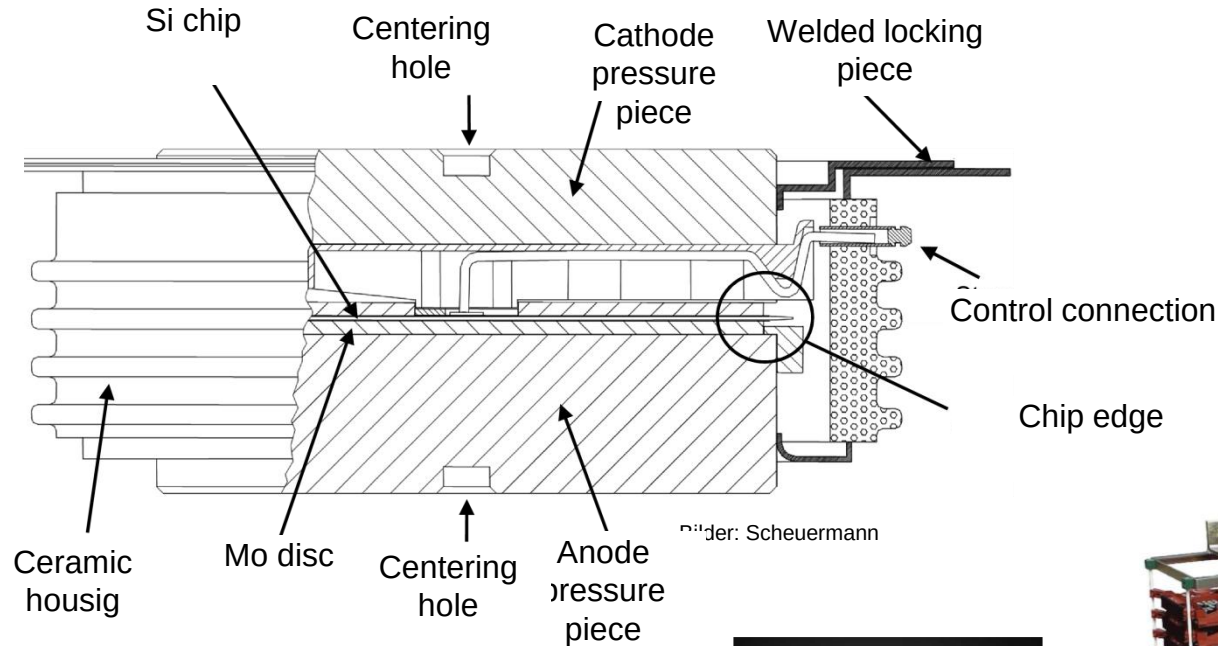


Image: Scheuermann

Advantages

- Hermetic encapsulation
- Double-sided cooling
- High reliability due to the possible omission of a molten connection

Disadvantages

- High demands on material tolerances (surface quality)
- High assembly effort
- No dielectric isolation

Differences in thermal expansion behavior cause increased thermo-mechanical stress in modules with copper base plates.

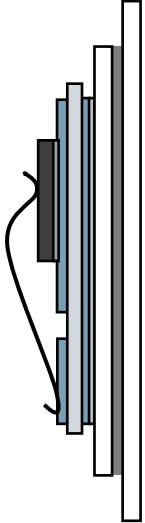
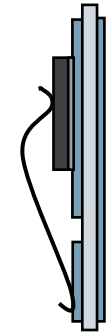
	Advantages	Disadvantages
With Baseplate 	- Improved and more homogeneous heat distribution over the chip surface (vs. Heat-spreader) - Modules with base plates are more robust in assembly	- Limitation of the substrate size - Higher static thermal resistance - Complication of the base plate soldering process - Increased thermo-mechanical stress due to CTE (Coefficient of Thermal Expansion) differences
Without Baseplate 	- No limitation of substrate size, which allows complex circuits on a single substrate - Reduced costs due to lower manufacturing effort and saving of base plate material - Increased reliability	Increase of short-term thermal impedance, as a lower heat capacity is available for buffering short-term overloads



Bild: Semikron SEMiX2s

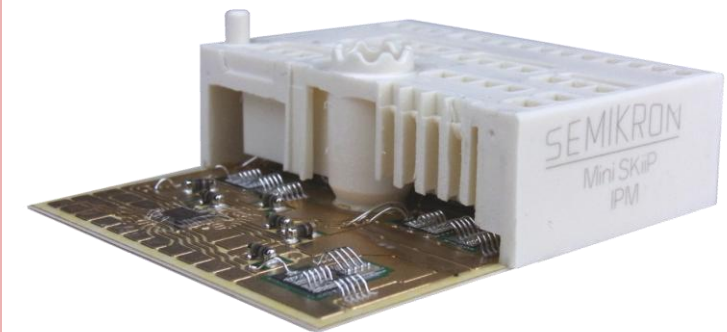


Bild: Semikron MiniSkiIP IPM

Process-accompanying quality assurance along the entire production line leads to high quality standards and lower production scrap rates.

System diagnosis

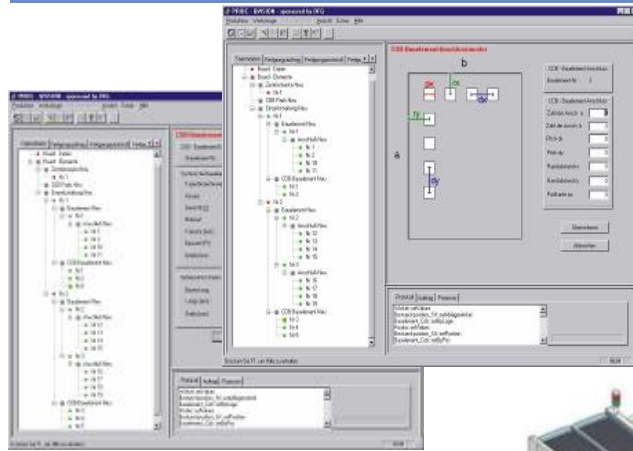


Bild: FMMSMT

Electrical testing

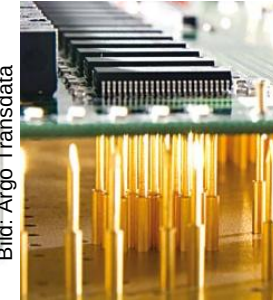


Bild: Argo Transdata

In Circuit Test

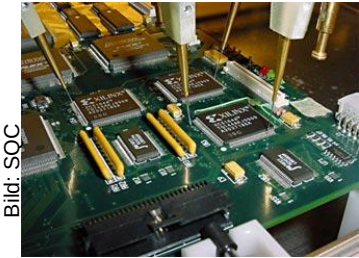
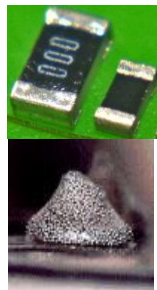


Bild: SQC

Flying Probe



Incoming goods inspection

Bilder: FAPS



Process analysis

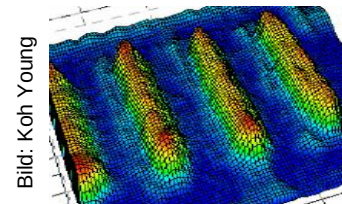
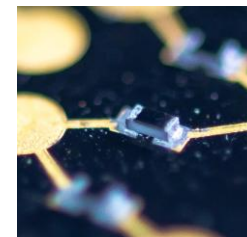
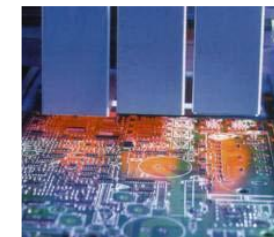


Bild: Koh Young

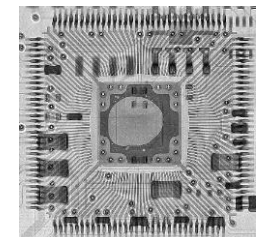
Solder paste inspection (SPI)



Component Inspection



Automatic optical inspection (AOI)



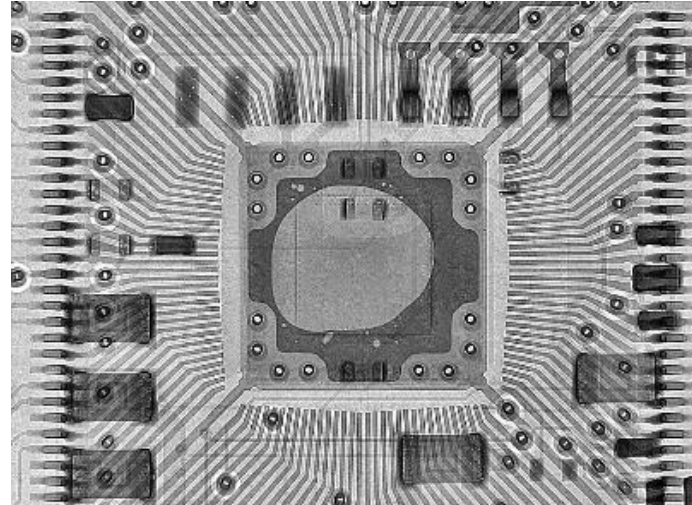
X-Ray Analysis (AXI)

Non-destructive testing by visual inspection methods can be performed inline as part of the process as well as separately.

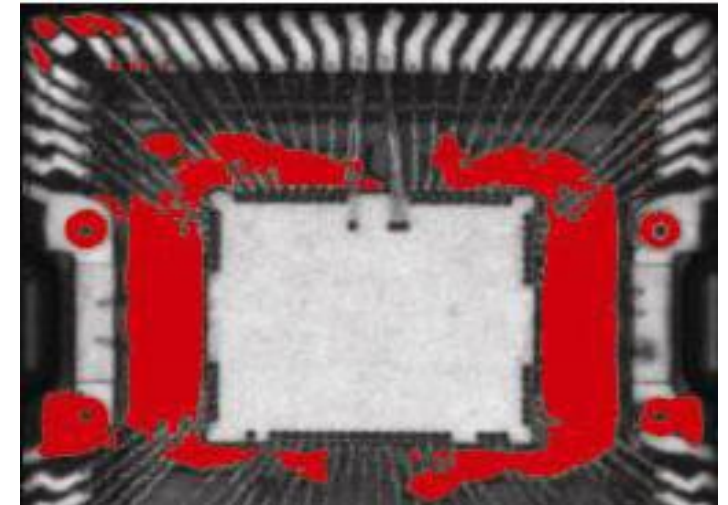
Automatic Optical Inspection (AOI)



X-Ray Analysis (AXI)



Scanning Acoustik Microscopy (SAM)



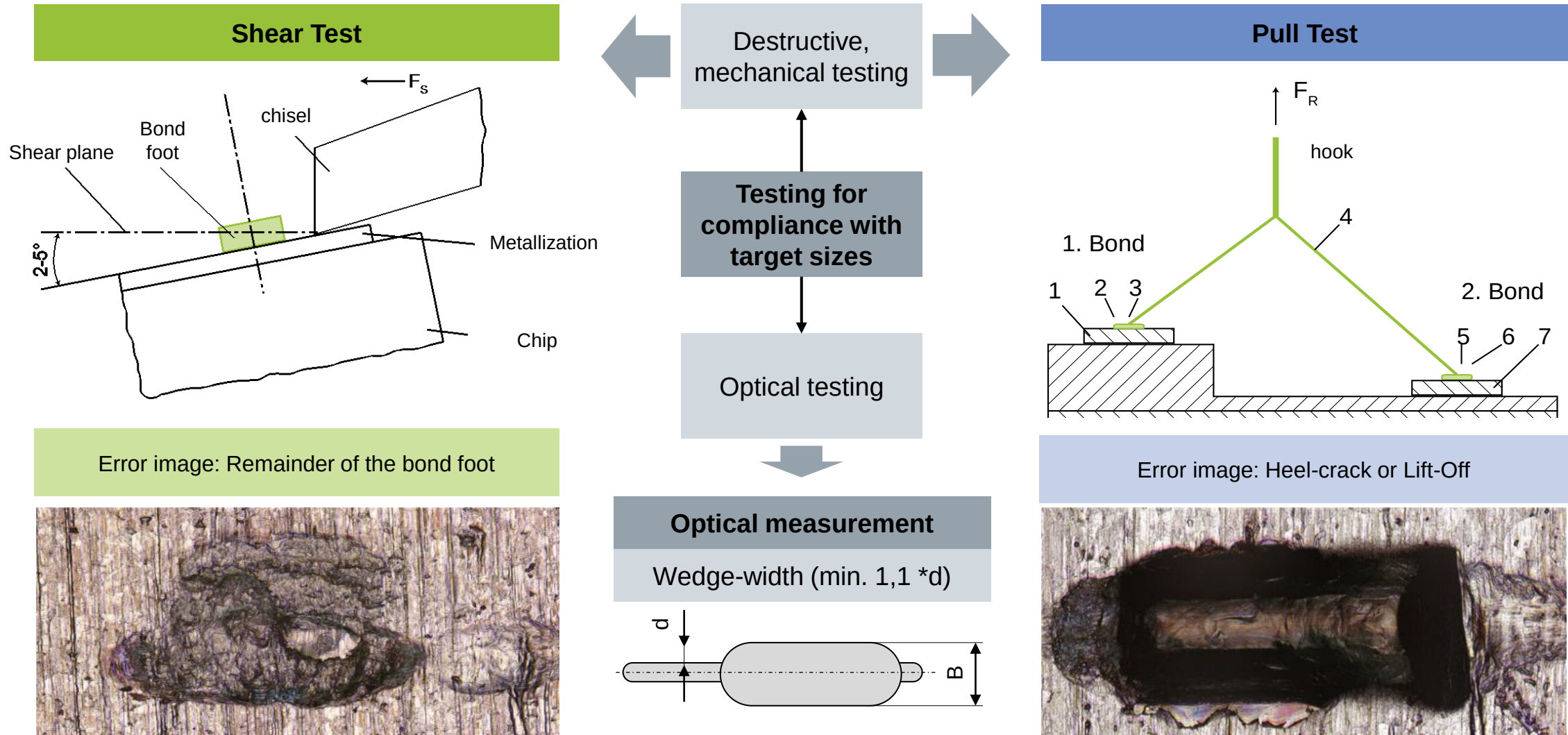
Typical applications and examples:

- Incorrect component position
- Missing component
- Wrong component
- Poor or missing solder joints

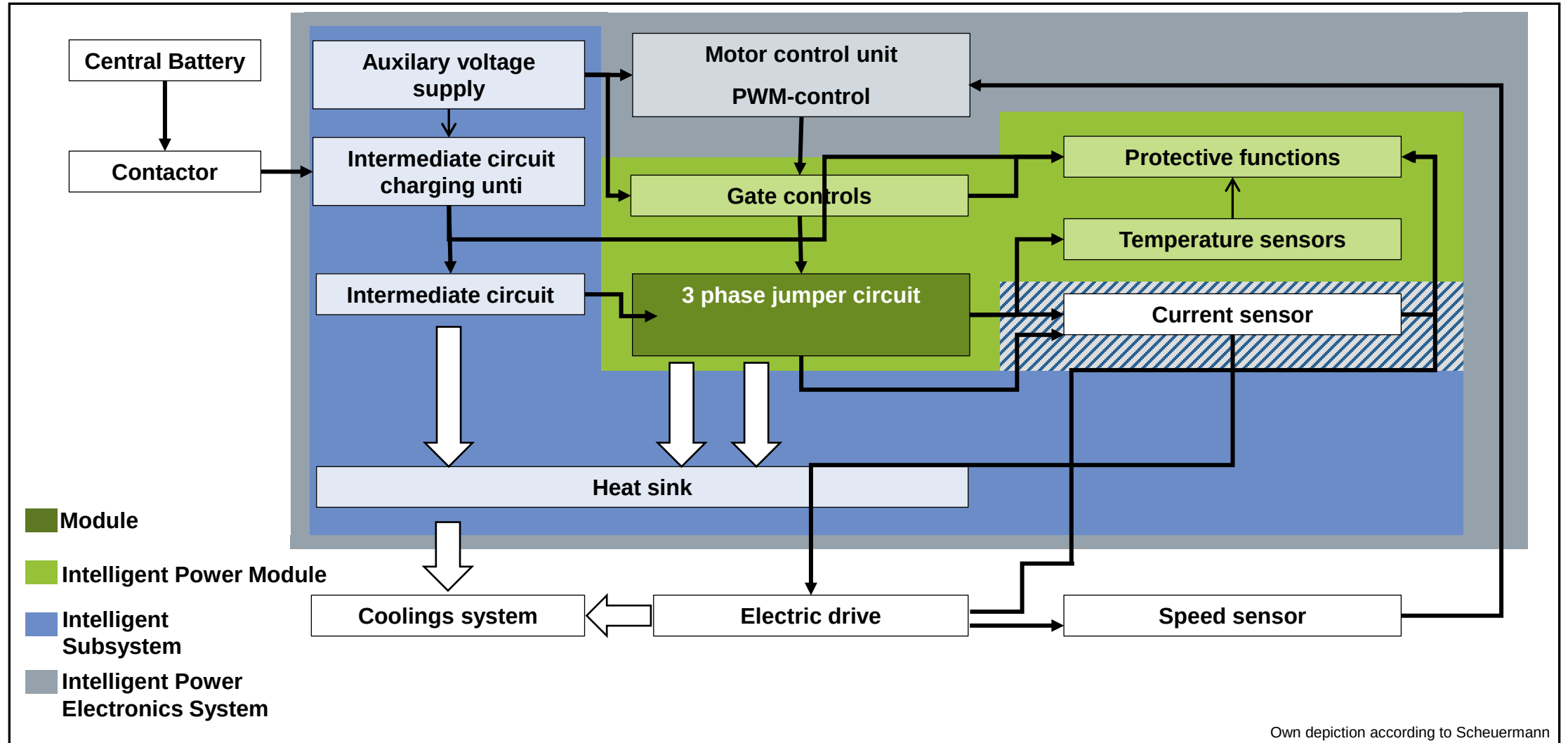
- Inspection of hidden solder joints
- Air gaps (voids) in different layers or solder joints
- Solder cracks
- Incomplete solder melting

- Analysis of optically non-transparent samples
- Detection of microcracks
- Identification of inclusions
- Delamination (e.g., in the area of the DBC)

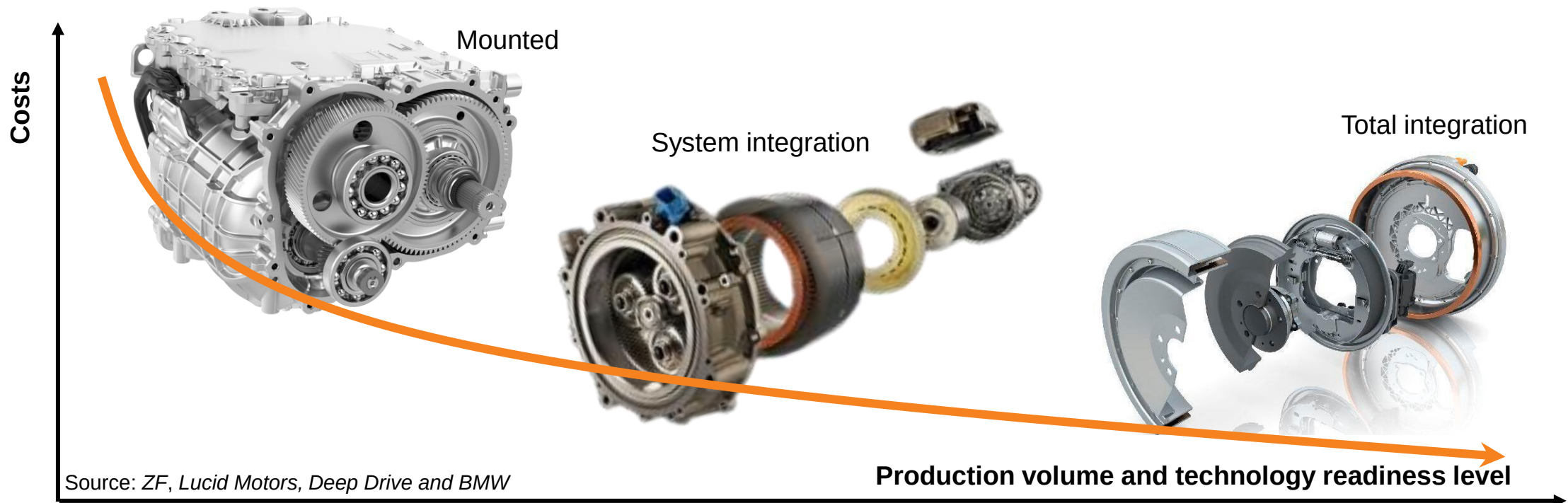
In addition to optically non-destructive testing methods, destructive shear and pull tests enable a meaningful quality assessment.



Through the module-adjacent implementation of additional functions, a power module can be expanded into a power electronics system, as is the case with electric vehicles.



Power electronics systems for the drive train illustrate the increased technological requirements, which arise from comprehensive spatial and functional integration in the modules.



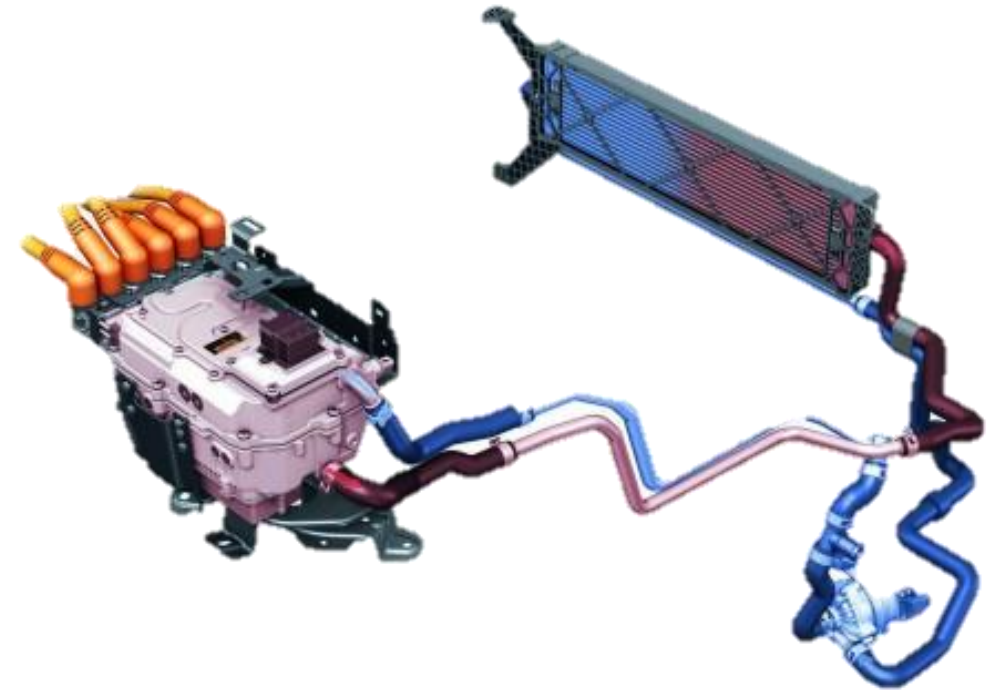
Integration of E-Machine, Transmission, and Power Electronics enables:

- Reduction of production costs
- Reduction of space requirements
- Scaling of the drive train
- Use of a single shaft
- Common cooling circuit
- Increasing efficiency
- Reduction of interfaces
- Automated assembly



Power electronics must be adapted to the electric drives just like the control and cooling systems.

- Introduction to power electronics
- Drive components in an electric vehicle
- Assembly and interconnection technology
- Module assembly
- **Thermal management**
- Reliability



Source: Audi AG

Every power module is part of a subordinate system – thermal management has to be considered in every system.

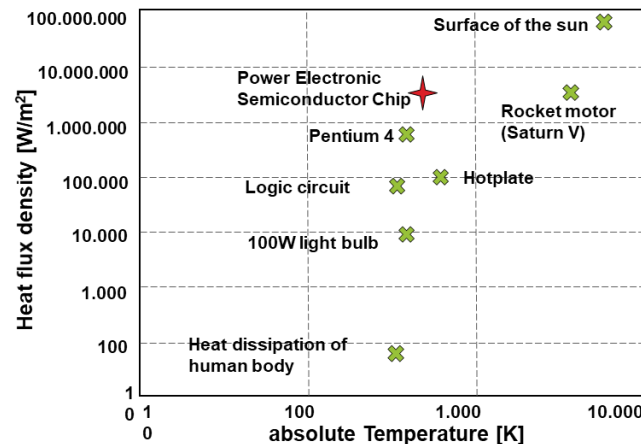


System Integration

Subsystem

Assembly Group (with housing)

Power Module Packaging



Controlling

Passive components

Semiconductor technology

Housing

Circuit carrier (PCB and DCB)

System integration

Assembly and interconnection technology

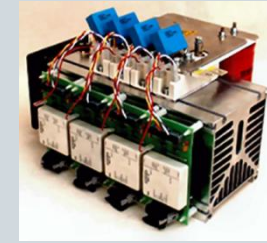
Thermal Management

Simulation and reliability prediction

To operate semiconductor components effectively, a system design with efficient thermal management is necessary.

Air cooling

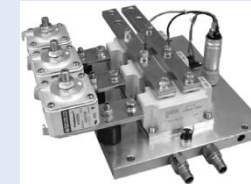
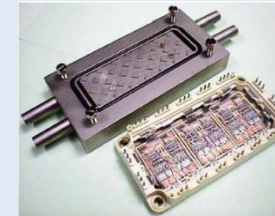
- Natural convection
- Forced convection



Bilder: Semikron, Aavid Thermalloy

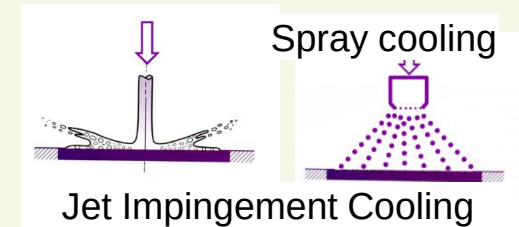
Liquid cooling

- Water, Water/Glycol
- Oil or inert liquids



Cooling by phase change

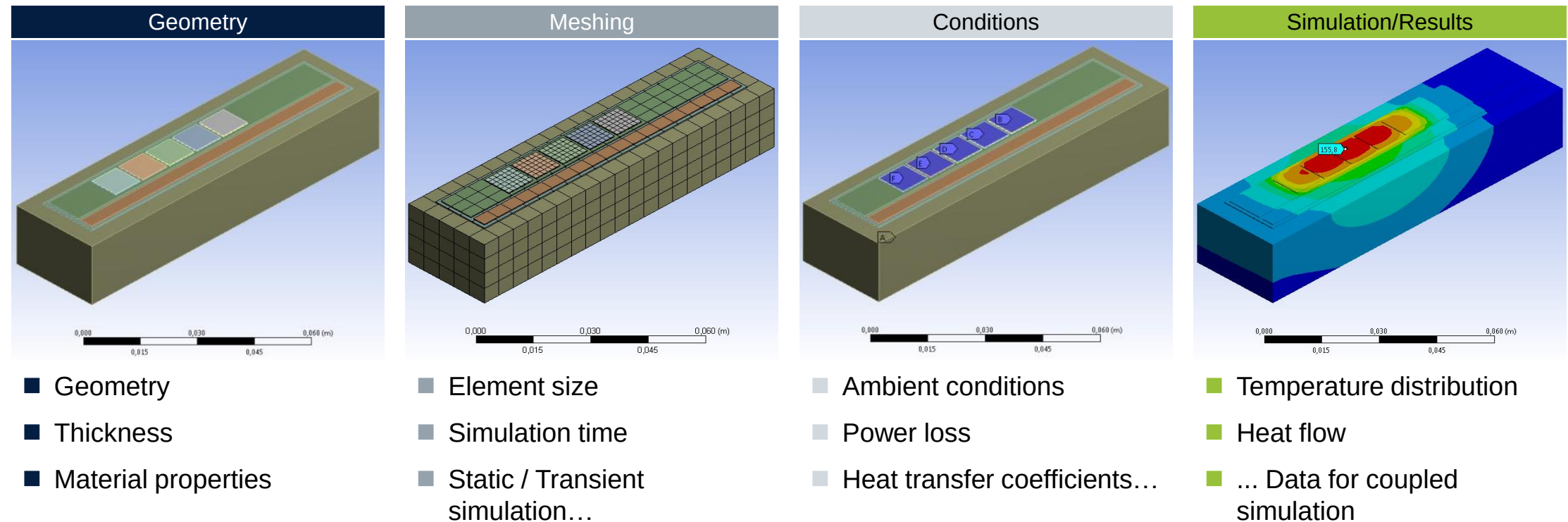
- Boiling cooling
- Heat pipes
- Cooling by spraying



Thermal simulation is an indispensable element in determining temperature development and distribution.

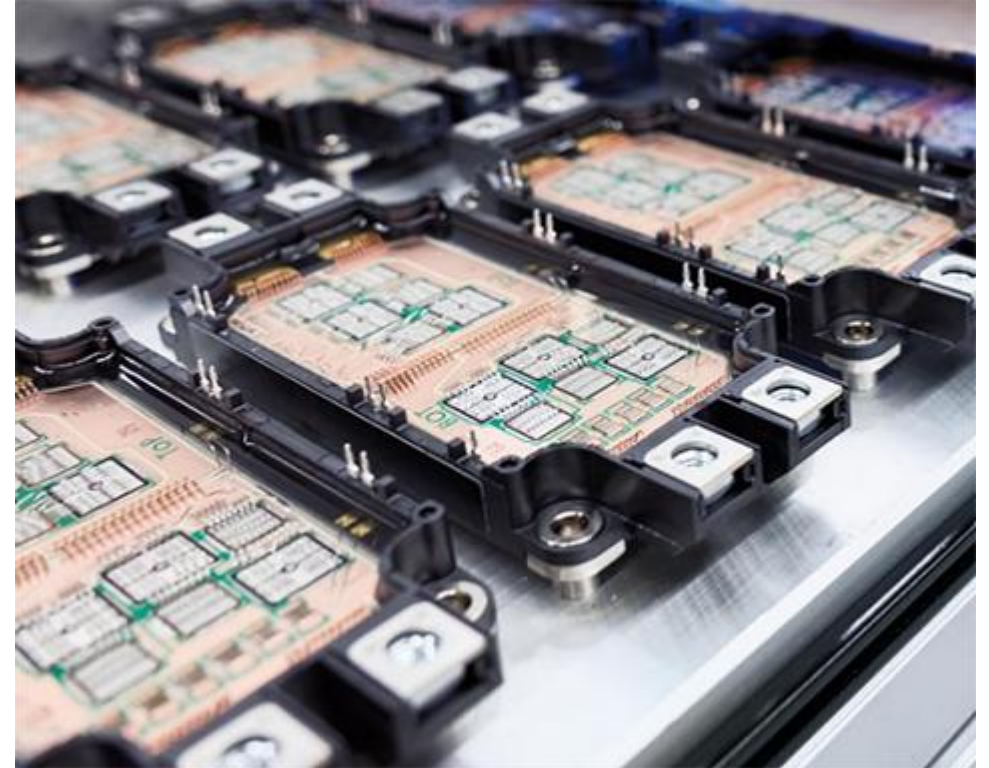
A complete thermal calculation is very complex, as it has to take into account exact geometrical relationships as well as heat transfer by conduction, convection, and radiation. The individual transfer rates are again dependent on the adjacent system, such as the installation situation and the installation space. However, often adequate approximations can be made through sensible simplification.

Process sequence



Power electronics must be adapted to the electric drives just like the control and cooling systems.

- Introduction to power electronics
- Drive components in an electric vehicle
- Assembly and interconnection technology
- Module assembly and testing
- Thermal management
- **Reliability**

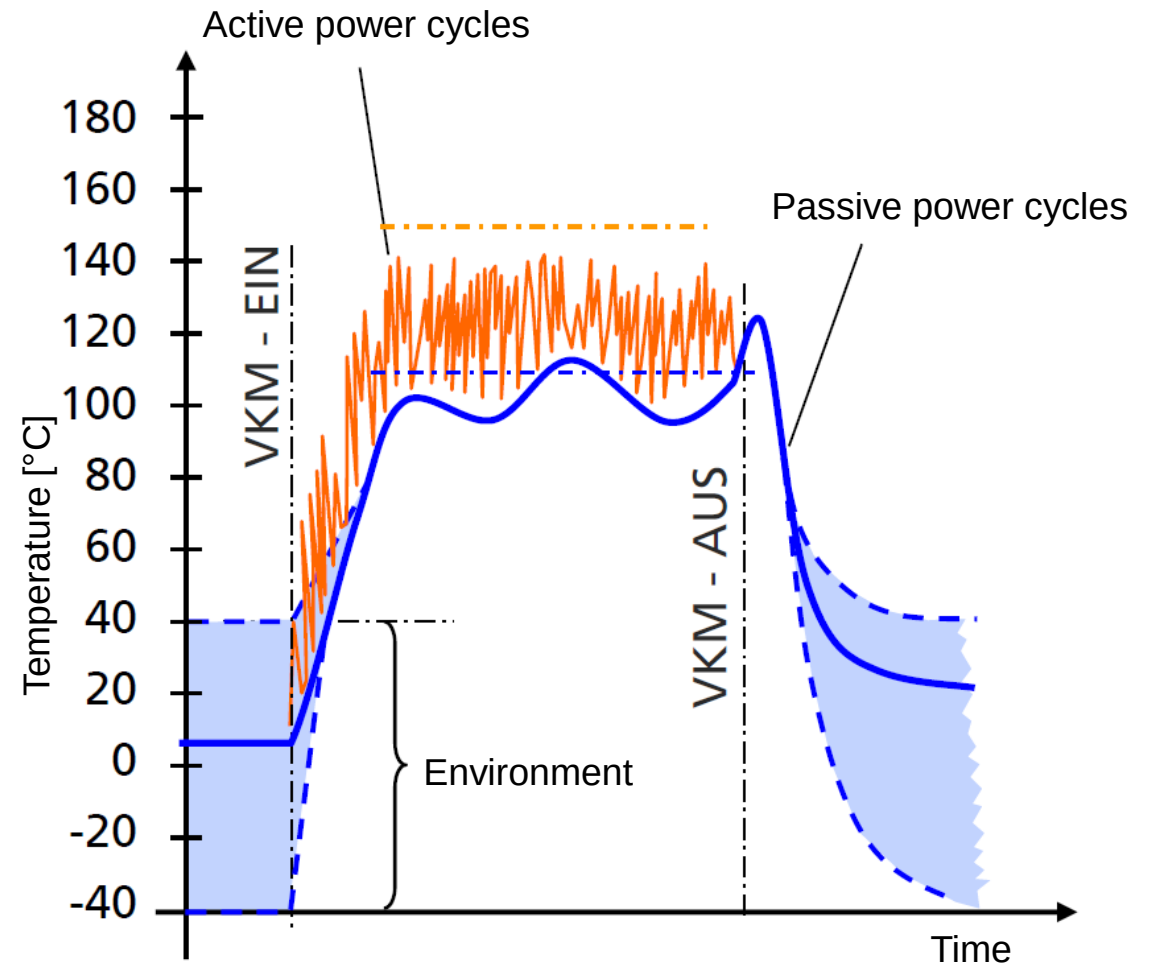


Source: Danfoss

Power loss through switching losses at the inverter causes temperature cycles over the entire product lifecycle.

Typical requirement profile of a Hybrid car:

Property	Requirement	Unit
Lifetime	15	Years
Operating hours	8,000	Hours
Distance driven	300,000	Km
Passive temperature cycles	10,000	#
Active temperature cycles	3-10 mio	#



To assess the reliability of connections and their durability, both passive and active environmental tests are performed.

Passive environmental tests

Damp Heat:

85° C, 85 % RH

100 | 250 | 500 (h)

Dry Heat:

200° C

100 | 200 (h)

Thermal Shock:

-40° C bis + 125° C

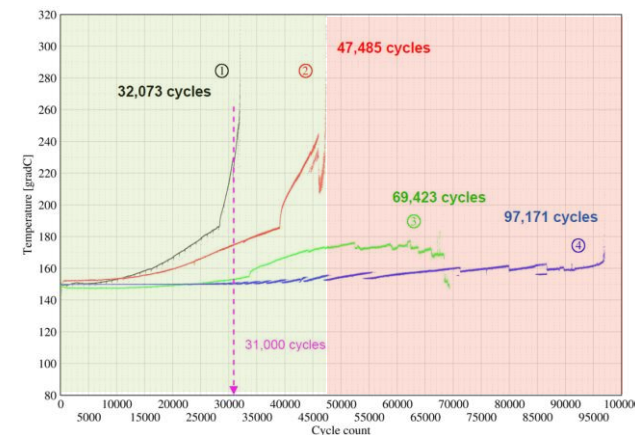
100 | 350 | 500 | 1000
(Cycles)



Active power cycling



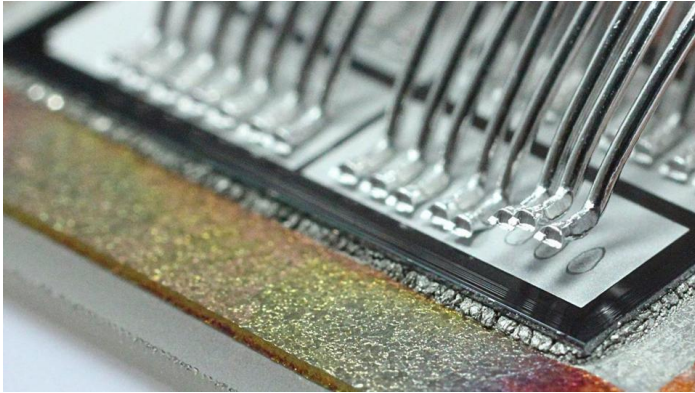
More realistic evaluation of connection quality through Active Power Cycling Possibility to combine with passive thermal cycling tests



Depending on the testing strategy, power modules can be stressed with a varying number of cycles. The choice of testing strategy decisively determines the maximum achievable cycle count.

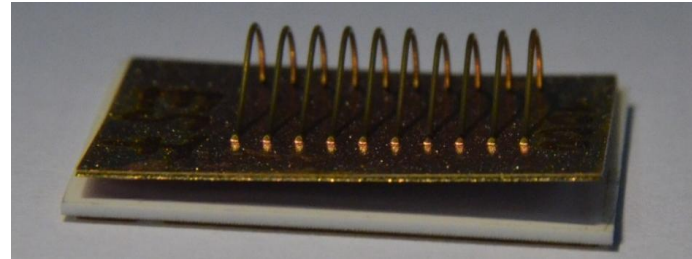
Active and passive stresses on semiconductor modules can be used to deliberately produce various failure mechanisms.

Degradation of chip metallization and bond wire lift-off



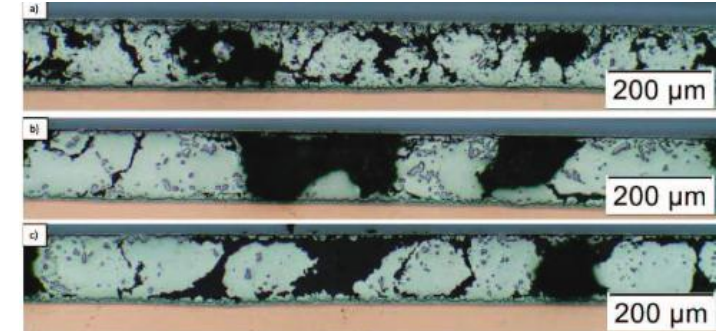
- CTE differences of the materials
- Material properties
- Surface quality
- Contamination by magnetic fields
- Vibration
- Bond parameters

Damage to insulated substrate (DCB)

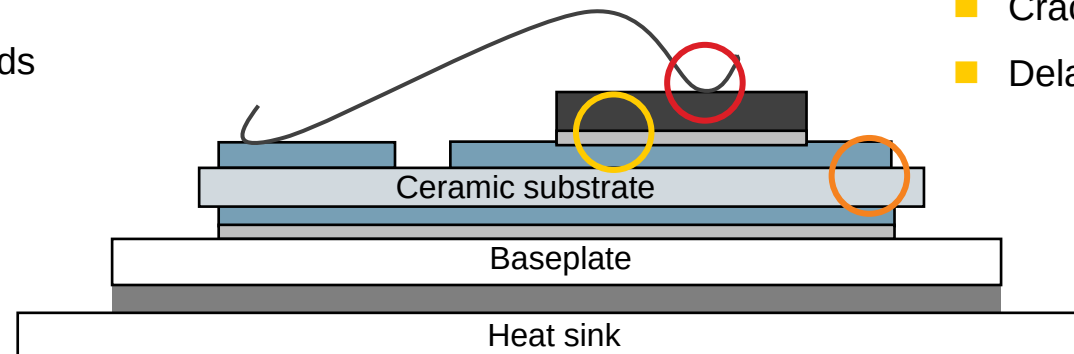


- Vertical cracks in substrate materials
- Corrosion
- Recrystallization of the metal layer
- Thin chip connection layer

Fatigue of solder layer



- CTE differences of the materials
- Material parameters
- Growth of intermetallic phases
- Cracks along intermetallic phases
- Delamination of chip metallization



Source: Fraunhofer IISB

In the future development, higher temperatures are just as necessary as higher reliability to save weight and increase performance.

Source: Semikron		Until 2014	Today	Future
Temperature limits		T_a 80 T_c 125 T_j 150 °C	T_a 80 T_c 125 T_j 175 °C	T_a 100 T_c 150 T_j 200 °C
Technologies	a) Die Attach	Soldering: $T_s = 180-220$ °C	Soldering: $T_s = 180-220$ °C Ag-Diffusionssintering	Transient liquid phase bond Ag-Diffusionssintering
	b) Housing Material	$T_{max} = 140^\circ\text{C}/10.000\text{h}$ $180^\circ\text{C}/1.000\text{h}$	$T_{max} = 140^\circ\text{C}/10.000\text{h}$ $180^\circ\text{C}/1.000\text{h}$	$T_{max} = 165^\circ\text{C} / 10000\text{h}$ $205^\circ\text{C} / 1000\text{h}$
	c) Encapsulation	$T_{max} < 150^\circ\text{C}$	$T_{max} < 175^\circ\text{C}$	$T_{max} > 200^\circ\text{C}$
	d) Wire Bonding	Standard Al Wire	Improved bond geometry, partially Cu-wires on Bond Buffers	Cu-Wire, Cu-/Al-Ribbons on Bond Buffer
Reliability	a) Power cycling	x1	x5	x25
	b) Temperature shock	$-40/+125^\circ\text{C} >$ >1000 Cycles	$-40/+125^\circ\text{C} > 1000$ Cycles, improved base plate soldering, design without baseplate	$-40/+150^\circ\text{C} > 1000$ Cycles, AlSiC / AlN-DCB Design without baseplate

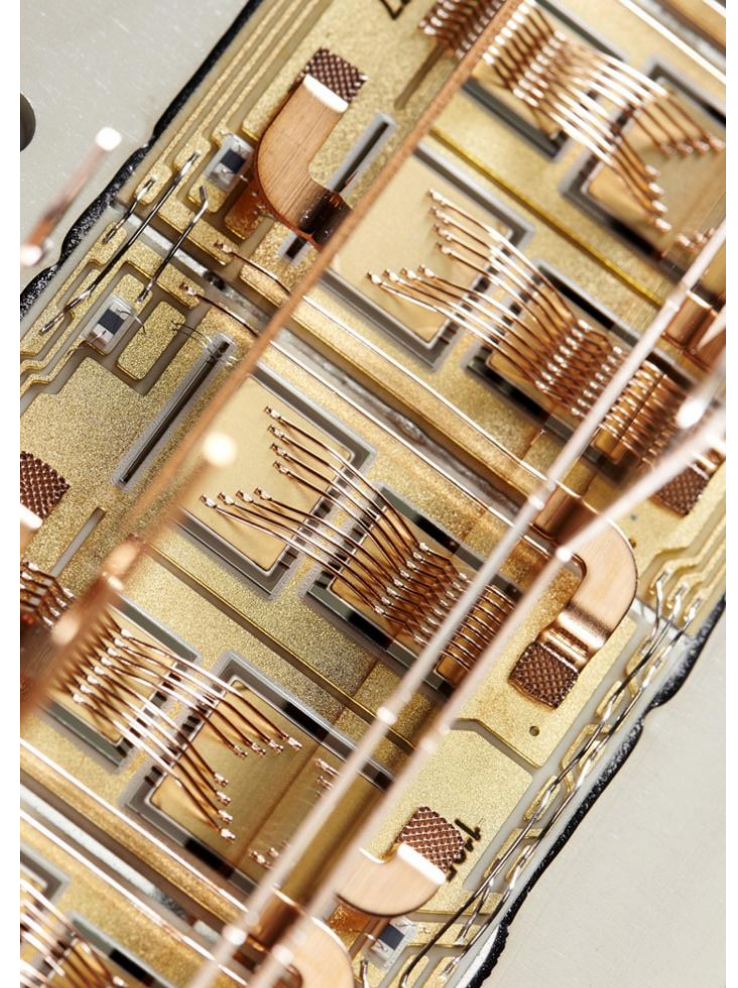
Therefore, future challenges in the manufacturing of modules for power electronics lie in contact-making and processing.

Contacting of power electronic components

- Automated process engineering for silver sintering, diffusion soldering, and vacuum processes
- Adapted wire bonding procedures and alternative contact-making possibilities for semiconductors
- Robust connection and integration of signal electronics to power electronics

Processing of power electronic assemblies:

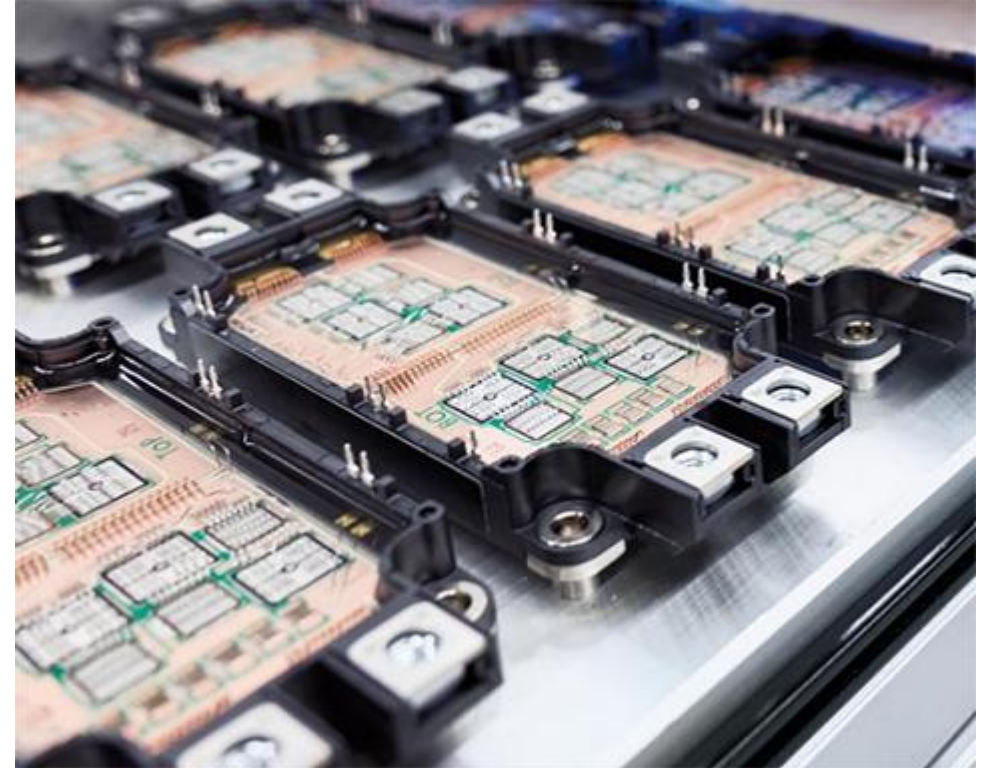
- Assembly concepts for components of thermal management
- Secure manufacturing processes for thick copper-clad substrates
- Targeted integration of heat removal structures
- Assembly procedures for three-dimensional construction of high-integration systems



Danfoss Bond Buffer with heavy copper wire bonds

Power electronics must be adapted to the electric drives just like the control and cooling systems.

- Introduction to power electronics
- Drive components in an electric vehicle
- Assembly and interconnection technology
- Module assembly and testing
- Thermal management
- Reliability
- **Power electronics research at Institute FAPS**

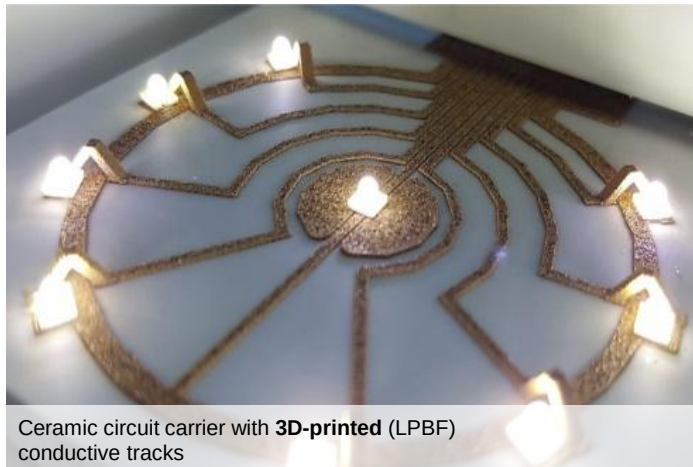


Source: Danfoss

The Institute FAPS is doing research on all relevant technologies for increasing efficiency and reliability in power electronics modules.

Substrate manufacturing

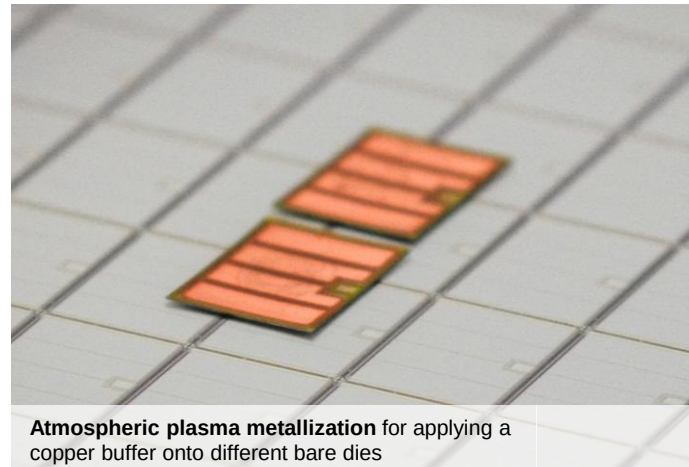
- Additive Manufacturing with
 - Laser Powder Bed Fusion for ceramics metallization (DCB) and cooler manufacturing
 - Fused Filament Deposition for 3D-ceramics
- Functionalization of 3D-substrates for high-temperature applications



Ceramic circuit carrier with **3D-printed** (LPBF) conductive tracks

Assembly and interconnection technology

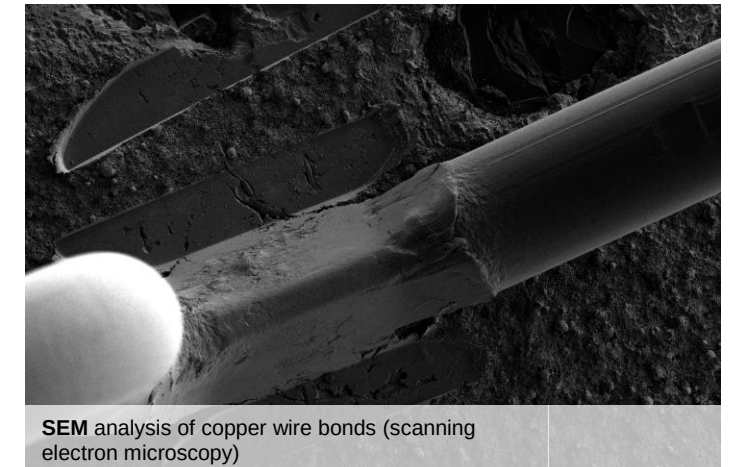
- Multi-material systems for diffusion soldering
- Silver and copper sintering
- Ultrasonic copper wire bonding
- Alternative metallization technologies for bare dies
- Hard encapsulation of power modules



Atmospheric plasma metallization for applying a copper buffer onto different bare dies

Characterization

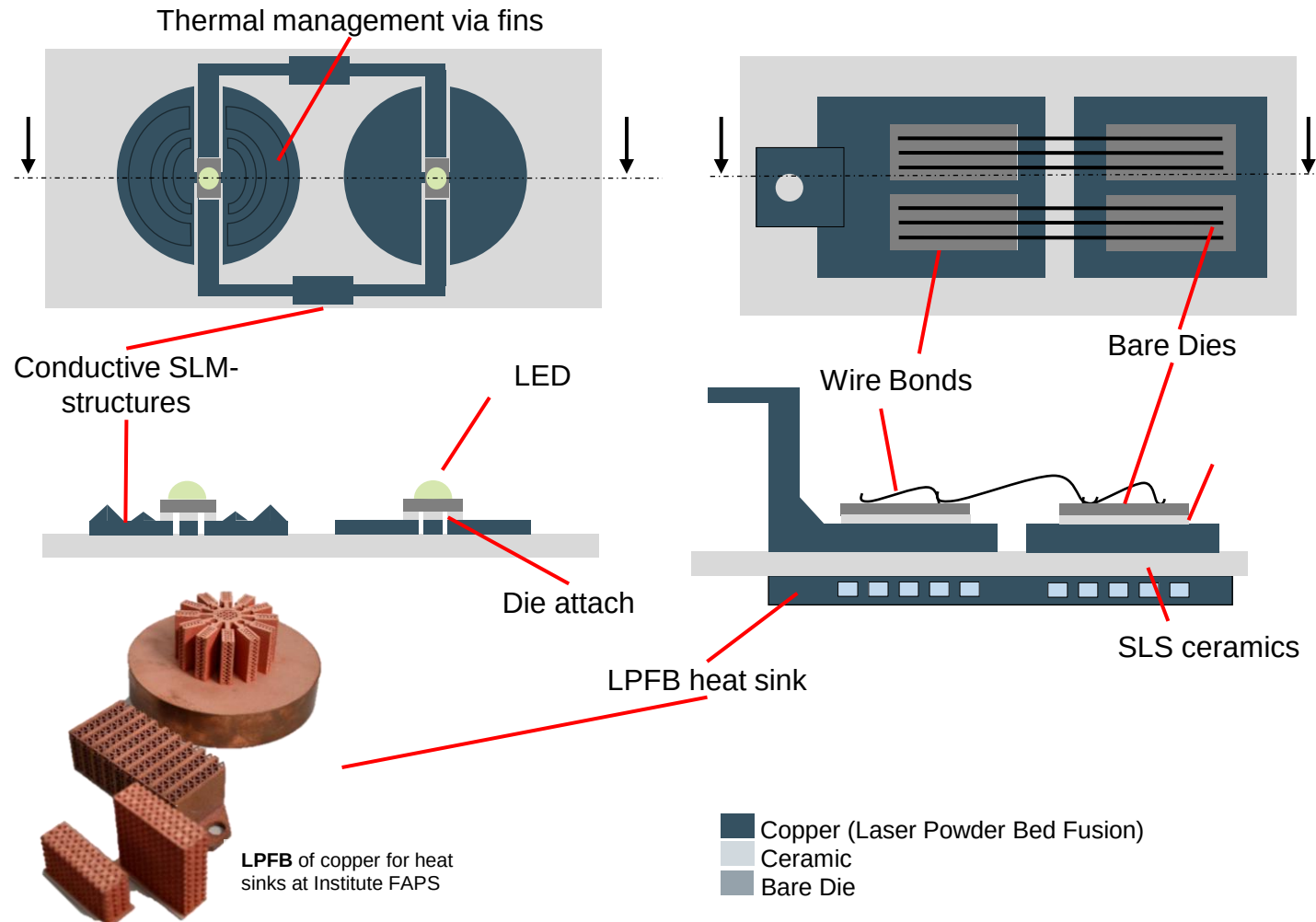
- Active and passive power cycling
- Identification of failure mechanisms
- Metallography and scanning electron microscopy



SEM analysis of copper wire bonds (scanning electron microscopy)

Additive manufacturing processes can be integrated into the production of power electronic components in the future.

Substrate manufacturing



- 3D design flexibility for electronic packaging.
- Customizable patterns and dimensions for design needs
- Innovative metal-ceramic connection manufacturing
- Advanced thermal management with specialized fin structures
- 3D-printed metallizations and copper-alloy heat sinks
- Efficient micro heat sinks for targeted hotspot cooling
- Simplified production process by removing conventional steps
- Enhanced heat transfer without thermal interface materials
- Material and weight savings

Innovative manufacturing technologies in assembly and interconnection technology significantly influence the performance of power modules.

Assembly and interconnection technology

Development of Systems and Processes for Low-Temperature Connection Techniques

- Sintering enables temperature-stable connections up to 950°C at process temperatures of approx. 260°C
- Homologous temperature is about 40% in comparison to standard solder and enables stable connections
- Copper sintering as a substitute for silver sintering
- Silver-silver diffusion bonding



SEHO Sinterpress at Institute FAPS

Increased reliability through Cu wire bonding

- Enhancement of metallization through chemical coating.
- More robust AVT due to the mechanical and electrical properties of copper
- Identification of process windows and qualification of connection quality



Plasma Coating Unit for power electronic bare dies at Institute FAPS

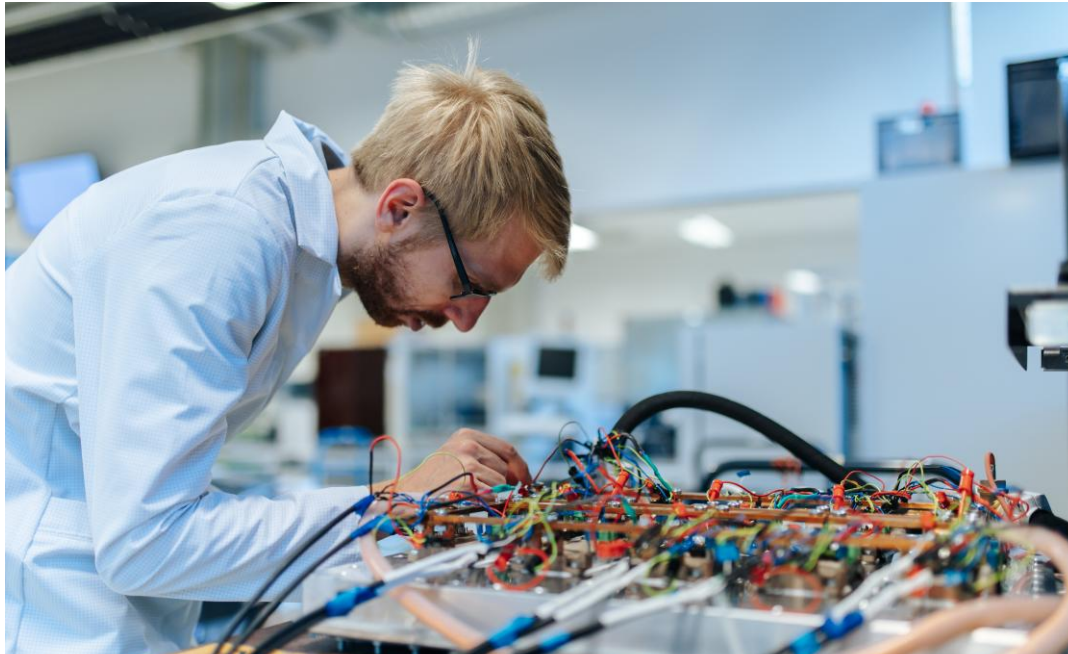
Picture: TF FAU | FATHER&SUN

To test power electronics and identify potential weaknesses active and passive power cycling as well as the characterization plays a huge part in research.

Characterization

Active & passive power cycling

- Testing of new materials (SiC) and testing strategies
- Production of specific system failures
- Thermal-shock, thermal treatments, etc.

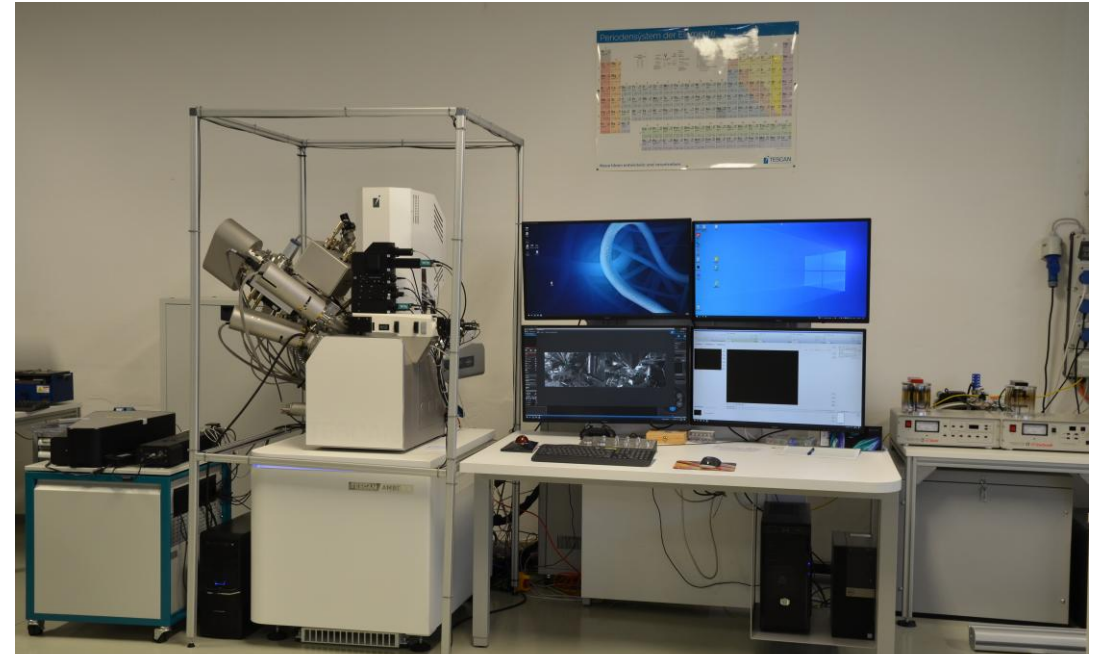


Active Power Cycling Set-Up at Institute FAPS

Picture: TF FAU | FATHER&SUN

Characterization via optical and destructive testing

- Scanning Electron Microscopy for identification of failure mechanisms on a microscopy level
- Laser scanning, optical microscopes, etc.



Scanning Electronic Microscopy with EDX, EBSD and Raman Spectroscopy for characterization on a microscopic level at Institute FAPS

Die Inhalte der Vorlesungseinheit können mit folgender Fachliteratur weiter vertieft werden:

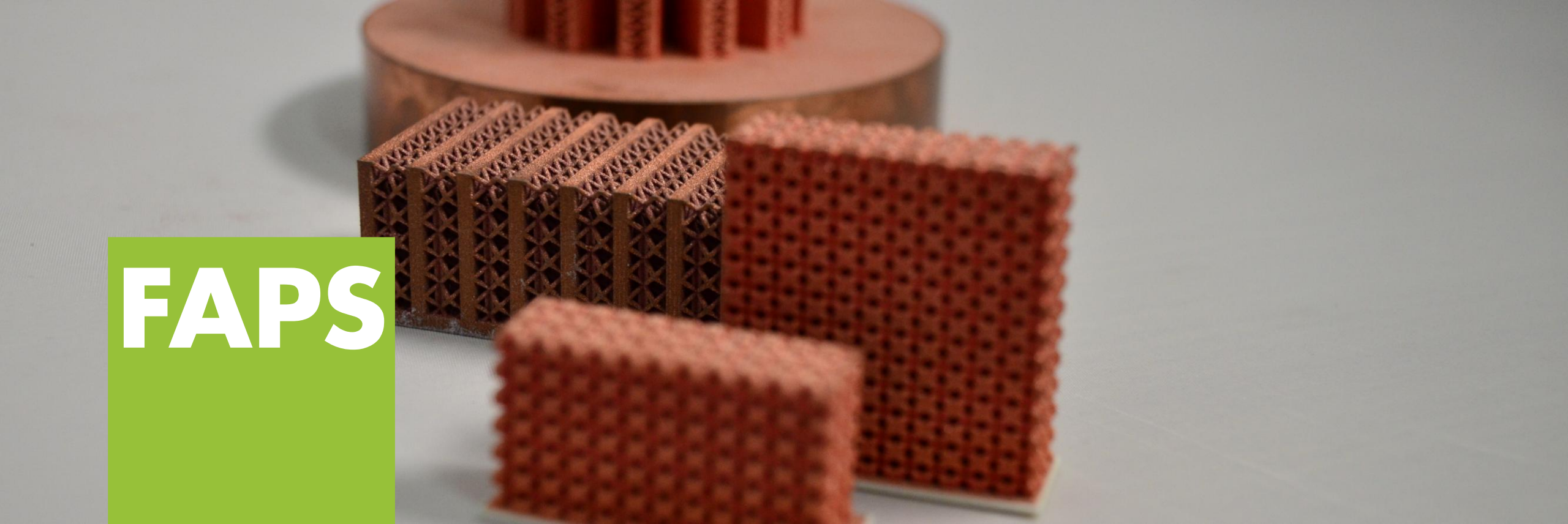
Lehrbücher

- FELDMANN K. (2009): Montage in der Leistungselektronik für globale Märkte. VDI-Buch. Springer, Berlin, Heidelberg
- SCHEUERMANN U. (2006): Aufbau- und Verbindungstechnik in der Leistungselektronik. In: Leistungselektronische Bauelemente. Springer, Berlin, Heidelberg
- KOMARNICKI P., HAUBROCK J., STYCZYNSKI Z.A. (2018): Elektrische Komponenten des E-Kfz. In: Elektromobilität und Sektorenkopplung. Springer Vieweg, Berlin, Heidelberg

Ergänzende Vorlesungen an der FAU

- Franke J. - Produktionsprozesse in der Elektronik (PRIDE 2)
- Scheuermann U. – Aufbau und Verbindungstechnik in der Leistungselektronik





FAPS

Prof. Dr.-Ing. Jörg Franke
Prof. Dr. rer. nat. Jens Fürst
Prof. Dr.-Ing. Florian Risch

Lehrstuhl für Fertigungsautomatisierung
und Produktionssystematik



Friedrich-Alexander-Universität
Technische Fakultät

DANKE